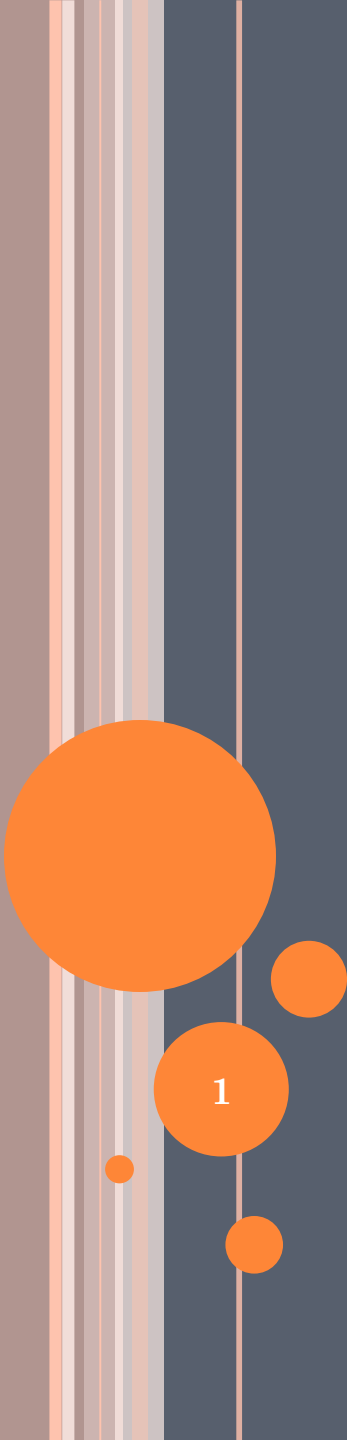


A decorative graphic on the left side of the slide. It features several vertical stripes in shades of orange, grey, and white. Overlaid on these stripes are several orange circles of different sizes. One large circle is positioned near the top, and a smaller circle below it contains the number '1'.

DEPLOY TWO TIER ARCHITECTURE IN AWS

1

OVERVIEW OF THE MODULE

This project is built on AWS Compute Services (AWS CCP) and teaches the basics of using cloud computing. It explains how computing works in the cloud and introduces important AWS services and ideas. Main topics are:

- The basics of cloud computing and computing services
- Virtual Machines and what they do in the cloud
- Amazon EC2 for virtual servers that can grow
- EC2 parts, types of instances, and naming rules
- Starting and running EC2 instances (for both Linux and Windows)
- Set up a secure web server (HTTPD) and SSH access
- Amazon ECS for managing containers
- AWS Elastic Beanstalk makes it easy to deploy applications
- Ideas for serverless computing
- AWS Lambda lets you run code without having to manage servers.

This module gives you the hands-on skills you need to deploy, manage, and scale applications in the AWS cloud environment using the Cyberarians infrastructure.

PROJECT OBJECTIVES AND DURATION

- Launch and configure a Linux-based EC2 instance for Cyberarians' cloud operations.
- Gain basic understanding of Virtual Machines, EC2, ECS, and serverless computing.
- Approximately 90 minutes.

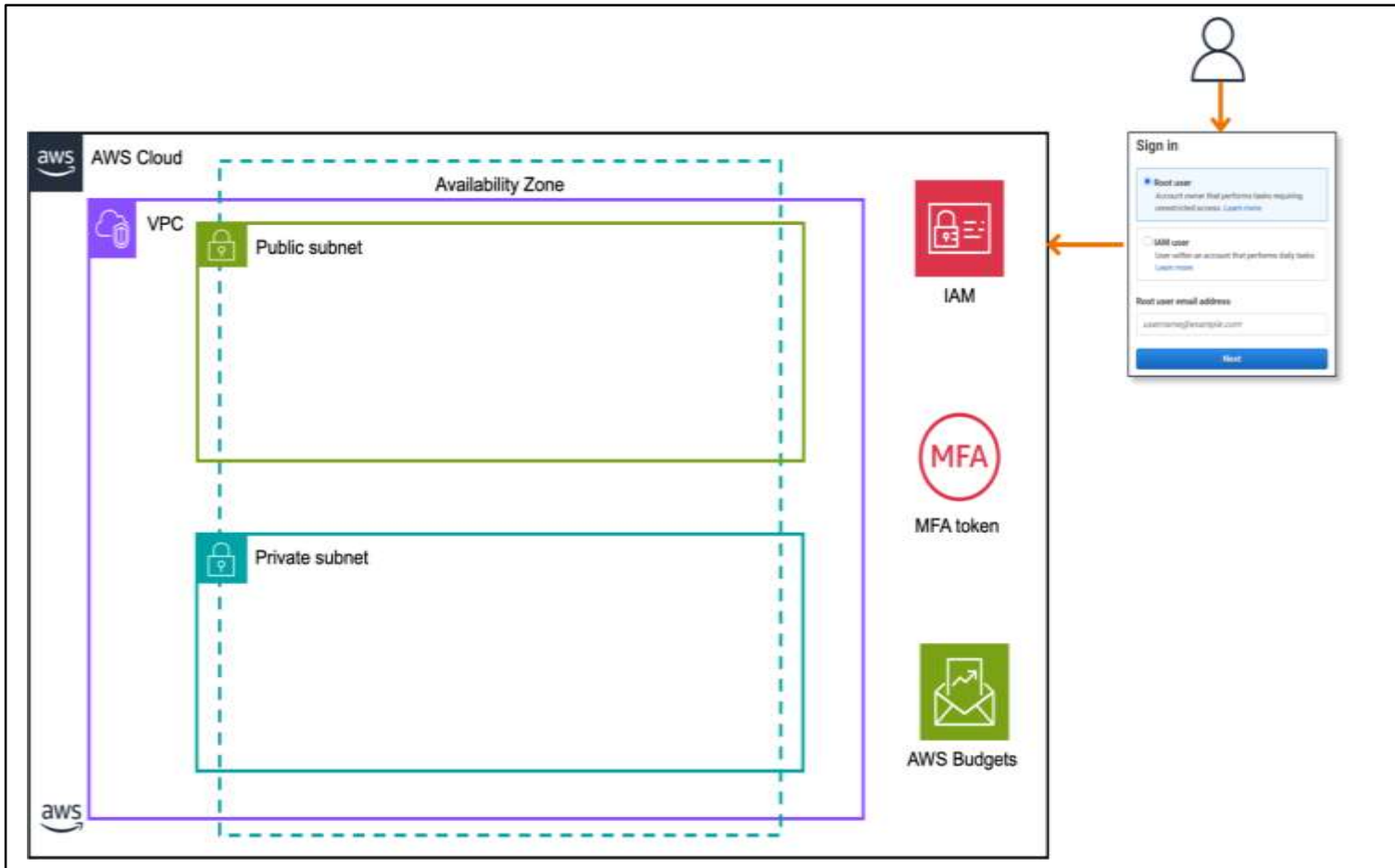
Start Project:

In this solution, you will walk through the step-by-step process of the tasks:

1. Launch a Linux-based EC2 instance using Amazon Linux 2 AMI.
2. Configure the instance with the necessary settings for web server operations.
3. Establish a secure SSH connection and perform basic network tests.

Let's start now!

TASK 1: SETTING UP AN AWS VIRTUAL PRIVATE CLOUD (VPC) WITH PUBLIC AND PRIVATE SUBNETS FOR CYBERARIANS



4/26/2026

AWS Cloud Security and Infrastructure
Implementation

Figure 1.1.1: Virtual Private Cloud

PROJECT OVERVIEW:

This project involves designing and deploying a Virtual Private Cloud (VPC) within the AWS environment for Cyberarians. The setup includes both public and private subnets to create a secure, isolated network for hosting applications and services.

Project Objectives:

- Build a VPC that ensures strong security and proper network isolation for AWS resources.
- Configure public subnets for internet-facing resources and private subnets for restricted internal components.
- Design the VPC architecture following best practices to support security, scalability, and efficient resource management.

STEP 1: PLANNING AND DESIGN

- **1. Understand CIDR Notation:**
CIDR notation defines an IP range using an address and a prefix (e.g., /16, /24). The prefix shows how many bits are used for the network—smaller values (like /16) provide more IP addresses, while larger values (like /24) provide fewer.
- **2. Plan Subnets and Sizes:**
- Identify how many subnets are needed and their sizes based on usage.
- Divide them into **public subnets** (internet access required) and **private subnets** (restricted access).
- Example design: Use a VPC CIDR block of **10.0.0.0/16**.
 - Public subnets: **10.0.0.0/24**, **10.0.1.0/24**
 - Private subnets: **10.0.2.0/24**, **10.0.3.0/24**
- This structure ensures proper IP distribution, scalability, and clear network segmentation.

STEP 2: CREATING THE VPC:

1. Access the AWS Console:

- Log in to your AWS account and navigate to the AWS Management Console.

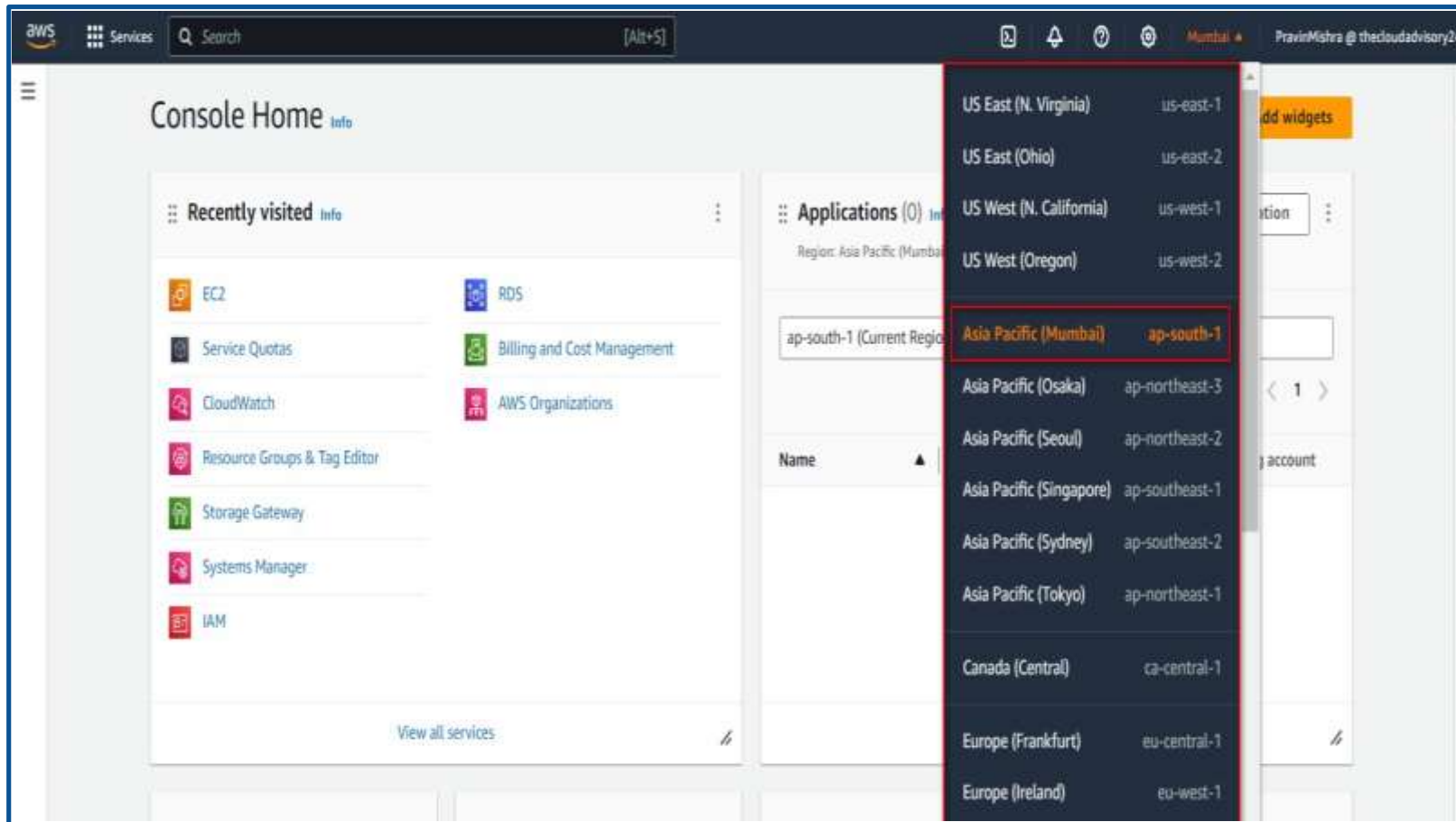
2. Choose the Desired Region:

- At the topmost corner of the AWS Management Console, select the region where you want to create your VPC. For instance, if you're selecting the Asia Pacific region, choose the appropriate region from the dropdown menu (*Figure 1.1.3: Region Selection*)

3. Go to the VPC Dashboard:

- In the AWS Management Console, locate the "Services" menu and select "VPC" under the "Networking & Content Delivery" section.

STEP 2: CREATING THE VPC:



4/26/2026

AWS Cloud Security and Infrastructure Implementation

Figure 1.1.3: Region Selection

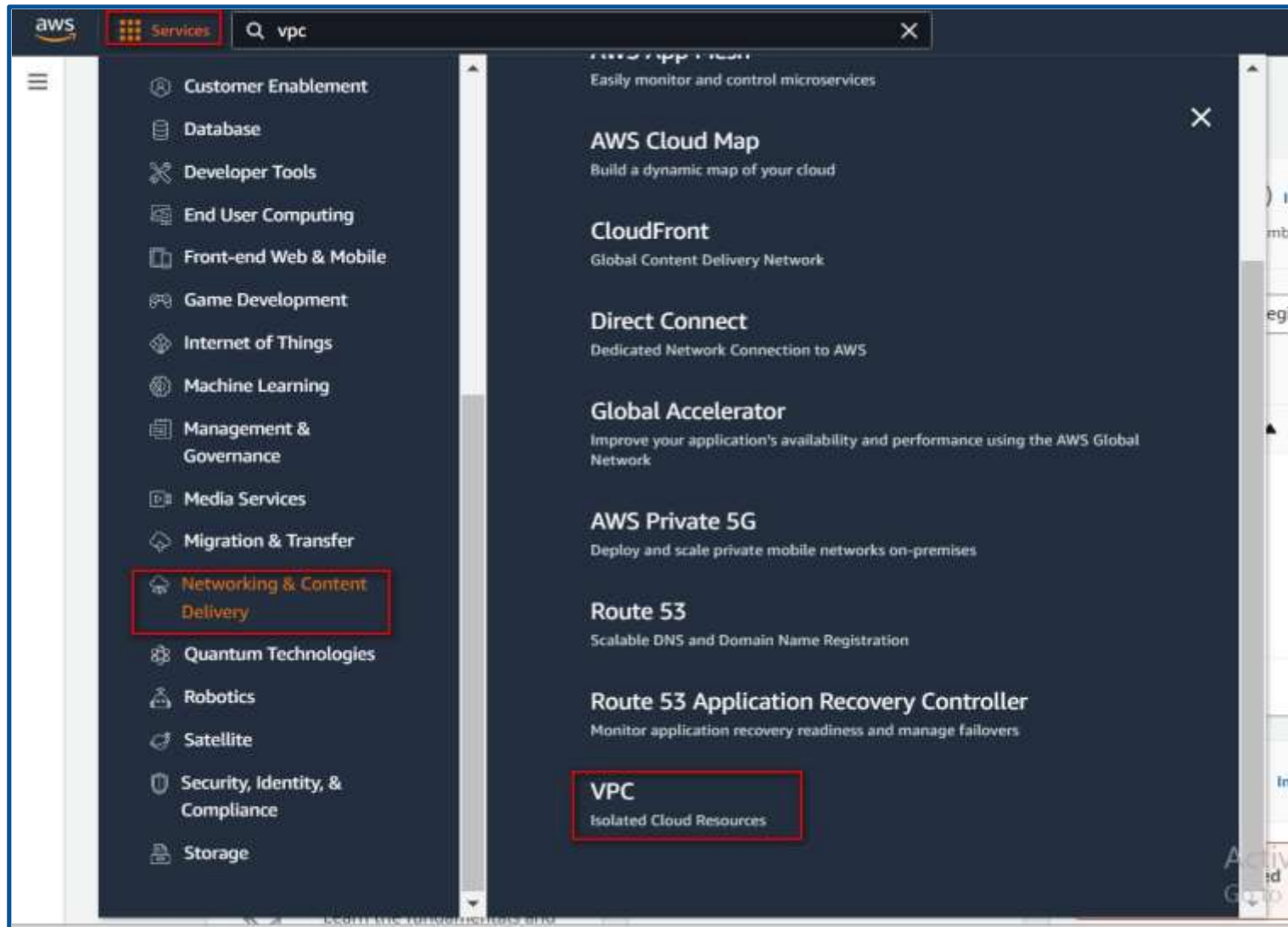


Figure 1.1.4: Select VPC

4. Create a New VPC:

- Click on "Create VPC" in the left sidebar of VPC Dashboard. (*Figure 1.1.5: Create VPC*)

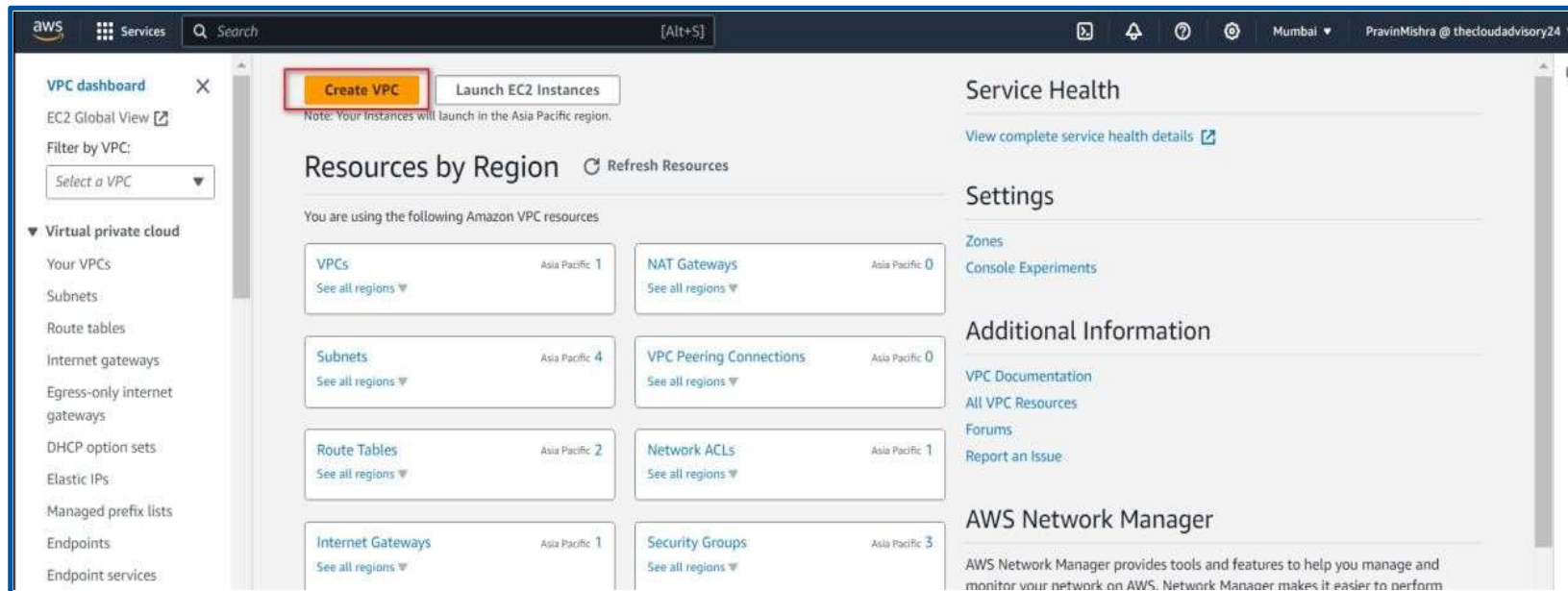


Figure 1.1.5: Create VPC

- In the resource to create option click on **"VPC only"** and provide the name of the VPC in the name tag.
- Enter the IPv4 CIDR block you previously selected in the earlier steps (for example, **10.0.0.0/16**) in the **"IPv4 CIDR block"** field. This defines the range of private IP addresses for your VPC's instances. (*Figure 1.1.6: VPC Settings*)

☒ VPC only ☐ VPC and more

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

Cyberarians-VPC

IPv4 CIDR block [Info](#)
☒ IPv4 CIDR manual input
☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR
10.0.0.0/16
CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)
☒ No IPv6 CIDR block
☐ IPAM-allocated IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block
☐ IPv6 CIDR owned by me

Tenancy [Info](#)
Default

VPC encryption control (\$) [Info](#)
Monitor mode provides visibility into encryption status without blocking traffic. Enforce mode prevents unencrypted traffic. [Additional charges apply](#)

☒ None ☐ Monitor mode ☐ Enforce mode

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key **Value - optional**

Q Name X Q Cyberarians-VPC X Remove tag

Add tag

You can add 40 more tags.

Cancel Preview code Create VPC

- VPC now creates and views the details in the VPC Dashboard. (*Figure 1.1.7: VPC Details*)

vpc-0aaea448590b81957 / Cyberarians-VPC

Details

VPC ID

 vpc-0aaea448590b81957

DNS resolution

Enabled

Main network ACL

[acl-07fcf352c8be0d338](#)

State

 Available

Tenancy

default

Default VPC

No

Block Public Access

 Off

DHCP option set

[dopt-016f09b04c7d6ac03](#)

IPv4 CIDR

10.0.0.0/16

DNS hostnames

Disabled

Main route table

[rtb-0ec2df85b00a254c1](#)

IPv6 pool

–

STEP 3: SETTING UP SUBNETS:

1. Create Public Subnets:

- In the VPC Dashboard, navigate to "Subnets" on the left sidebar and click on "Create subnet." (*Figure 1.1.9: Create Subnet*)

Public Subnet 1

- VPC: Cyberarians-VPC
- AZ: 1a
- CIDR: 10.0.1.0/24

Public Subnet 2

- AZ: 1b
- CIDR: 10.0.2.0/24

Private Subnet 1

- AZ: 1a
- CIDR: 10.0.3.0/24

Private Subnet 2

- AZ: 1b
- CIDR: 10.0.4.0/24

○ Public Subnet 1 and 2

NEW SUBNET

IPv4 CIDRs
10.0.0.0/16

Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of "Name" and a value that you specify.
1b
The name can be up to 256 characters long.

Availability Zone [info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
Europe (Stockholm) / eu-north-1a2 (eu-north-1b) ▼

IPv4 VPC CIDR block [info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must be within this block.
10.0.0.0/16 ▼

IPv4 subnet CIDR block
10.0.2.0/24 (16 IP)

▼ **Tags - optional**

Key	Value - optional	
Q Name	Q 1b	Remove

[Add new tag](#)
You can add 49 more tags.

[Remove](#)

[Add new subnet](#)

[Cancel](#) [Create subnet](#)

Figure 1.1.8: Subnet Settings

○ Private Subnet 1

Subnets > Create subnet

Associated VPC CIDRs

IPv4 CIDRs
10.0.0.0/16

Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of "Name" and a value that you specify.
Tc
The name can be up to 255 characters long.

Availability Zone [info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
Europe (Stockholm) / eu-n1-az1 (eu-north-1a)

IPv4 VPC CIDR block [info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must fit within this block.
10.0.0.0/16

IPv4 subnet CIDR block
10.0.3.0/24 (256 IPs)

▼ Tags - optional

Key	Value - optional	
Name	Tc	Remove

You can add 49 more tags.

4/26/2026

AWS Cloud Security and Infrastructure
Implementation

Figure 1.1.9: Subnet Settings

○ Private Subnet 2

4/26/2026

AWS Cloud Security and Infrastructure
Implementation

IPv4 CIDRs
10.0.0.0/16

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
1d
The name can be up to 256 characters long.

Availability Zone [info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
Europe (Stockholm) / eu-north-1b

IPv4 VPC CIDR block [info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must be within this block.
10.0.0.0/16

IPv4 subnet CIDR block
10.0.4.0/24 256 IPs

▼ **Tags - optional**

Key	Value - optional	
Q Name	Q 1d	Remove

[Add new tag](#)
You can add 49 more tags.

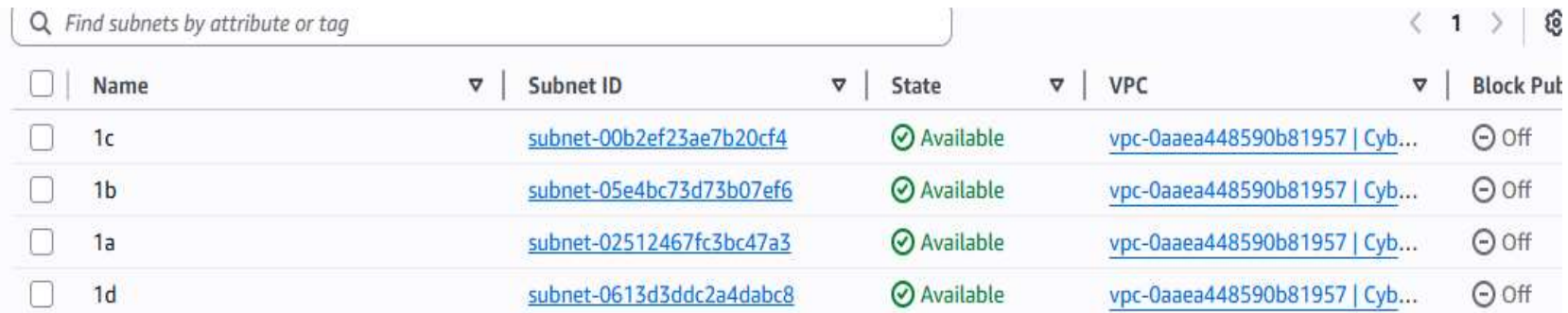
[Remove](#)

[Add new subnet](#)

[Cancel](#) [Create subnet](#)

Figure 1.1.10: Subnet Settings

- Get back to the VPC console, Select the Subnets and you will find the Subnets that you have created a while ago! (*Figure 1.1.11: Subnets of Cyberarians*)



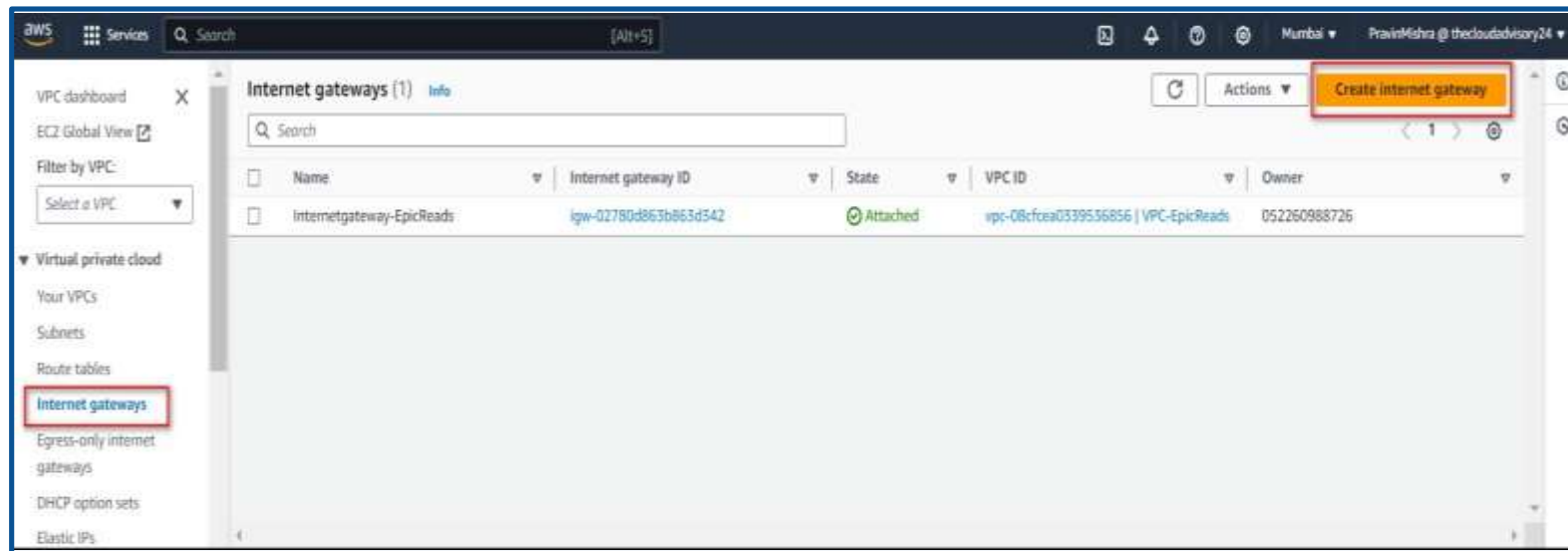
The screenshot shows the AWS VPC console interface. At the top, there is a search bar with the placeholder text "Find subnets by attribute or tag". Below the search bar is a table listing subnets. The table has columns for Name, Subnet ID, State, VPC, and Block Put. There are four subnets listed, all with a state of "Available".

☐	Name	Subnet ID	State	VPC	Block Put
☐	1c	subnet-00b2ef23ae7b20cf4	✓ Available	vpc-0aaea448590b81957 Cyb...	⊖ Off
☐	1b	subnet-05e4bc73d73b07ef6	✓ Available	vpc-0aaea448590b81957 Cyb...	⊖ Off
☐	1a	subnet-02512467fc3bc47a3	✓ Available	vpc-0aaea448590b81957 Cyb...	⊖ Off
☐	1d	subnet-0613d3ddc2a4dabc8	✓ Available	vpc-0aaea448590b81957 Cyb...	⊖ Off

STEP 4: CONFIGURING INTERNET ACCESS:

1. Set Up Internet Gateway (IGW) for Public Subnets:

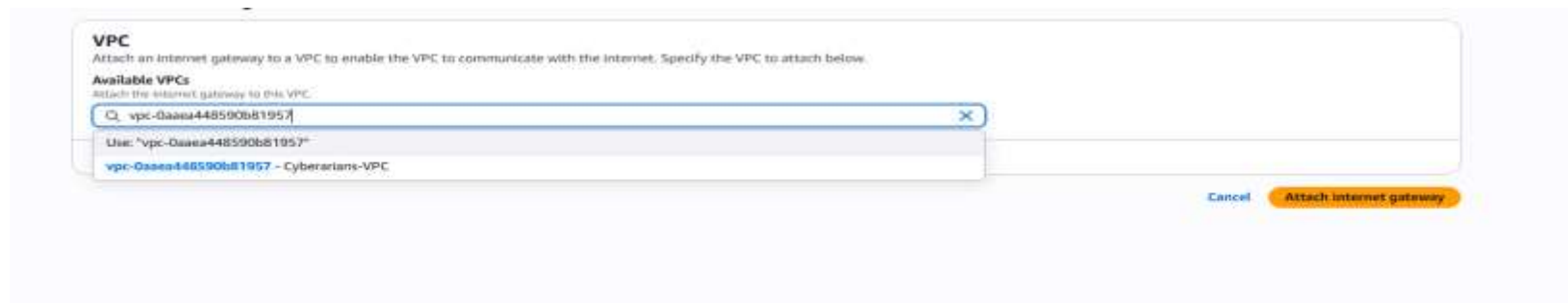
- In the VPC Dashboard, click on "Internet Gateways" in the left sidebar. (*Figure 1.1.13: Create Internet Gateway*)



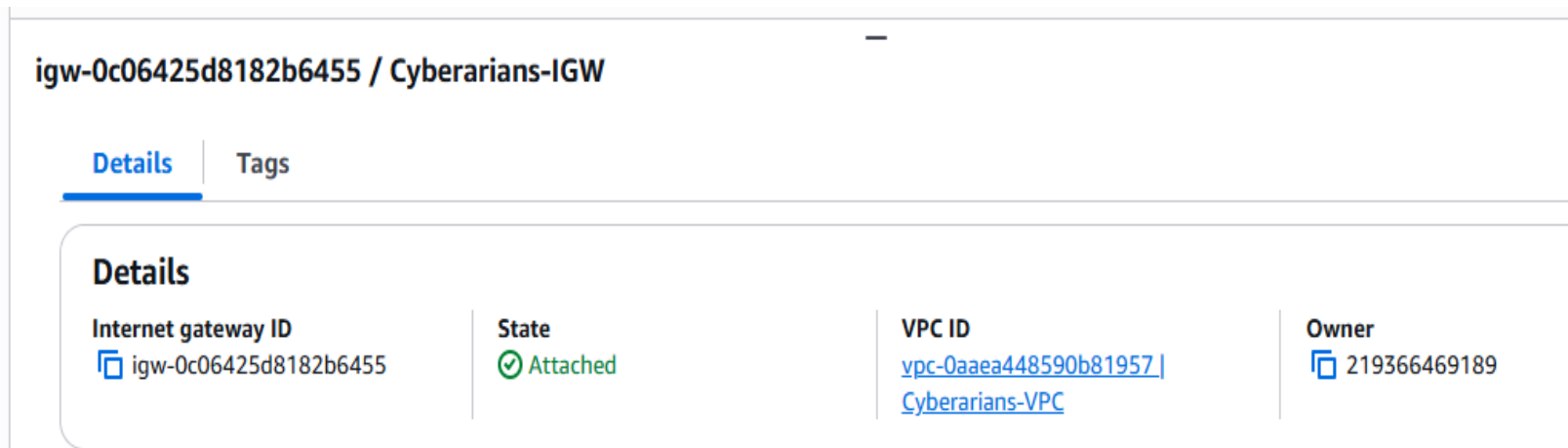
- Click on "Create internet gateway" and provide a name for the IGW. (*Figure 1.1.15: Internet Gateway Settings*)



- Select the newly created IGW and click on "Attach to VPC," then choose your VPC from the list. (*Figure 1.1.16: Attach to VPC*)



- Once VPC is attached, you can see the details in the internet gateways Tab. (*Figure 1.1.17: IGW Details*)



2. Update Route Tables for Public Subnets:

- VPC → Route Tables → Create
- Name: Public-RT
- VPC: Cyberarians-VPC

Destination	Target
0.0.0.0/0	Internet Gateway

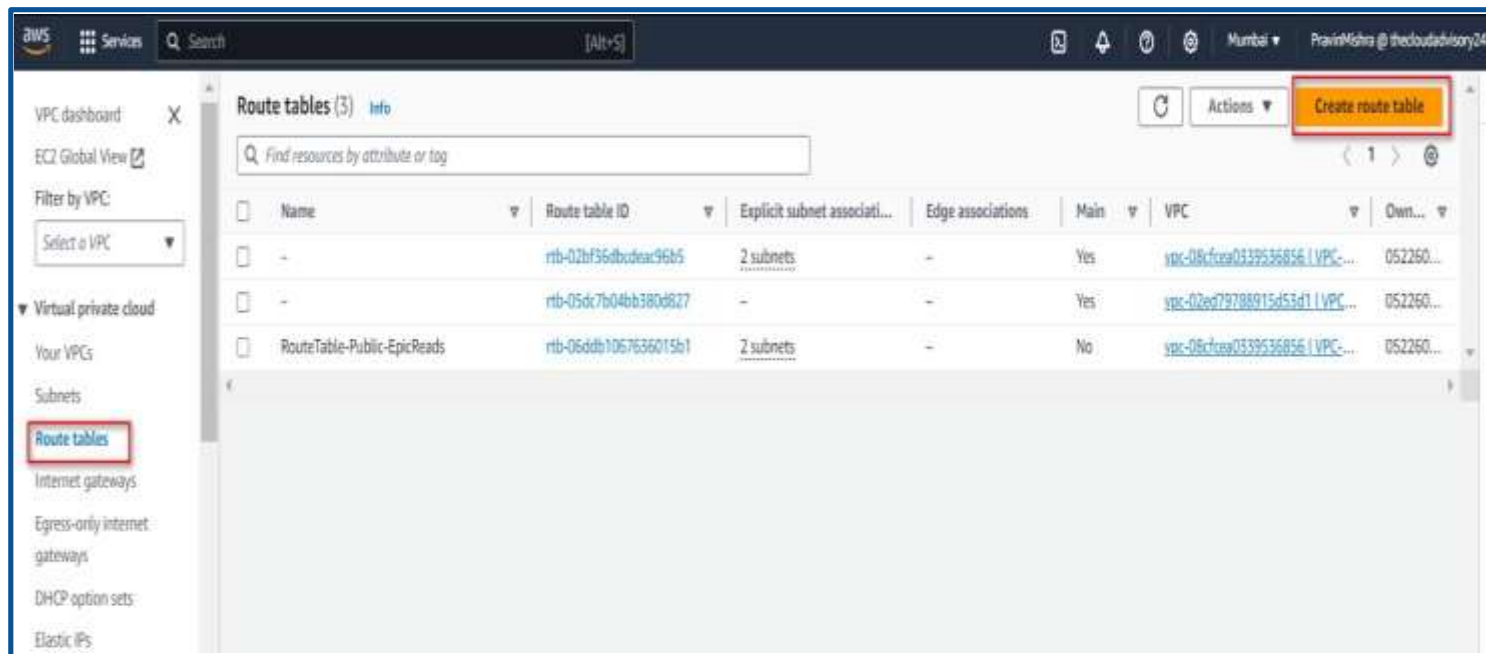


Figure 1.1.18: Route Tables

Associate: Attach:

- Public Subnet 1
- Public Subnet 2

✓ You have successfully updated subnet associations for rtb-03a32be202fa72775 / Public-RT.

Route tables (1/3) [Info](#)

Find route tables by attribute or tag

	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
☒	Public-RT	rtb-03a32be202fa72775	2 subnets	subnet-05e4bc73d73b07ef6 / 1b ✕ subnet-02512467fc3bc47a3 / 1a	No	vpc-0aaea448590b81957 Cyb...	219366469189
☐		rtb-0cf84df5f3af7a26e	-		Yes	vpc-0603634cbe9913884	219366469189
☐		rtb-0ec2df85b00a254c1	-		Yes	vpc-0aaea448590b81957 Cyb...	219366469189

Last updated: less than a minute

Figure 1.1.19: Route Tables

2. Private Route Table

Create another:

- Name: Private-RT

Important:

- DO NOT attach Internet Gateway here

Associate:

- Private Subnet 1
- Private Subnet 2



Figure 1.1.20: Route Tables

STEP 5: IMPLEMENTING NETWORK SECURITY:

1. Create Network Access Control Lists (NACLs):

- In the VPC Dashboard, click on "Network ACLs" in the left sidebar.
- Click on "Create network ACL."

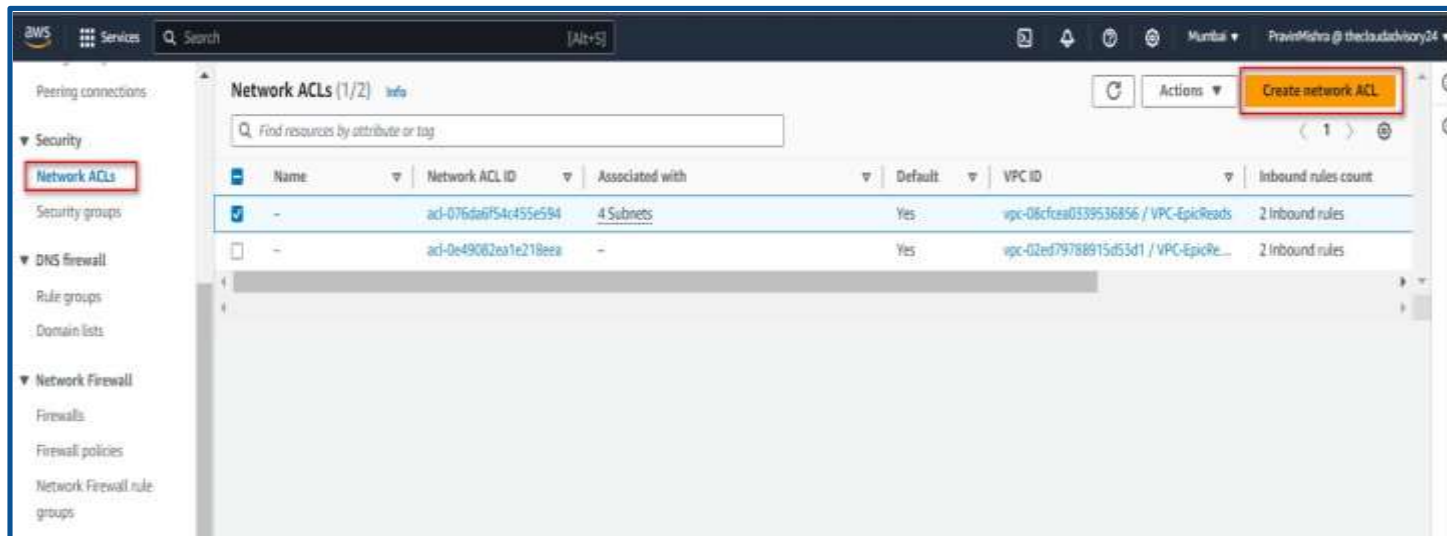


Figure 1.1.21: NACL details

- Configure inbound and outbound rules to control traffic.
- Remember that NACLs are stateless, so you'll need to define rules for both inbound and outbound traffic. (Figure 1.1.22: Edit Inbound Rules)

Edit inbound rules Info

Inbound rules control the incoming traffic that's allowed to reach the VPC.

Rule number Info	Type Info	Protocol Info	Port range Info	Source Info	Allow/Deny Info	
100	All traffic	All	All	0.0.0.0/0	Allow	<button>Remove</button>
101	SSH (22)	TCP (6)	22	0.0.0.0/0	Allow	<button>Remove</button>
102	HTTP (80)	TCP (6)	80	0.0.0.0/0	Allow	<button>Remove</button>
103	HTTPS (443)	TCP (6)	443	0.0.0.0/0	Allow	<button>Remove</button>
104	MS SQL (1433)	TCP (6)	1433	0.0.0.0/0	Allow	<button>Remove</button>
+	All traffic	All	All	0.0.0.0/0	Deny	

Add new rule Sort by rule number

Cancel Preview changes Save changes

Figure 1.1.22: Edit Inbound Rules

acl-087dd3c7a82a7bc76 / cybrarians-NACL

Outbound rules (2)

Edit outbound rules

Rule number	Type	Protocol	Port range	Destination	Allow/Deny
100	All traffic	All	All	0.0.0.0/0	✓ Allow
*	All traffic	All	All	0.0.0.0/0	✗ Deny

Figure 1.1.23: Edit outbound Rules

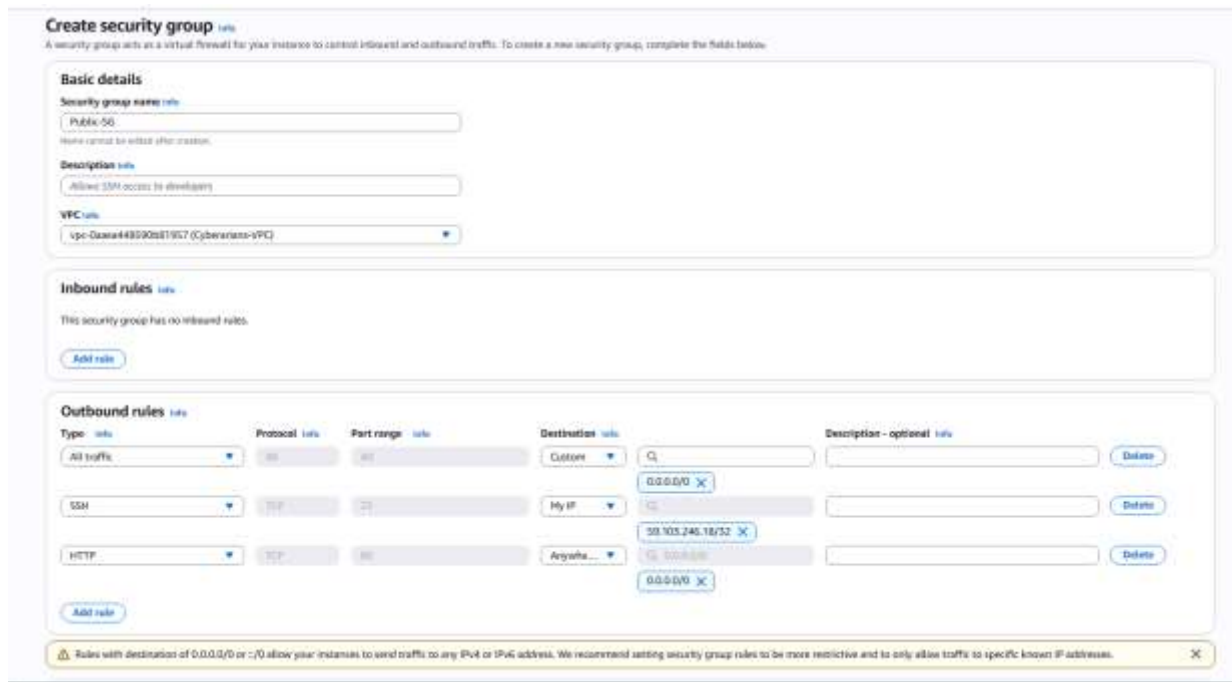
2. Create Security Groups:

- Go back to the VPC Dashboard and click on "Security Groups" in the left sidebar.
- Click on "Create security group."

Public EC2 Security Group

Name: Public-SG

- Rules: SSH (22) → your IP , HTTP (80) → 0.0.0.0/0



Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details [Info](#)

Security group name [Info](#)

Public-SG

Name cannot be edited after creation.

Description [Info](#)

Allowed SSH access to developers

VPC [Info](#)

vpc-0aaw448090819c7 (Cyberware-VPC)

Inbound rules [Info](#)

This security group has no inbound rules.

[Add rule](#)

Outbound rules [Info](#)

Type	Protocol	Port range	Destination	Description - optional	
All traffic	TCP	0	0.0.0.0/0		Delete
SSH	TCP	22	My IP		Delete
HTTP	TCP	80	Anywhere...		Delete

[Add rule](#)

Rules with destination of 0.0.0.0/0 or :::/0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific known IP addresses.

Figure 1.1.24: Public EC2 Groups

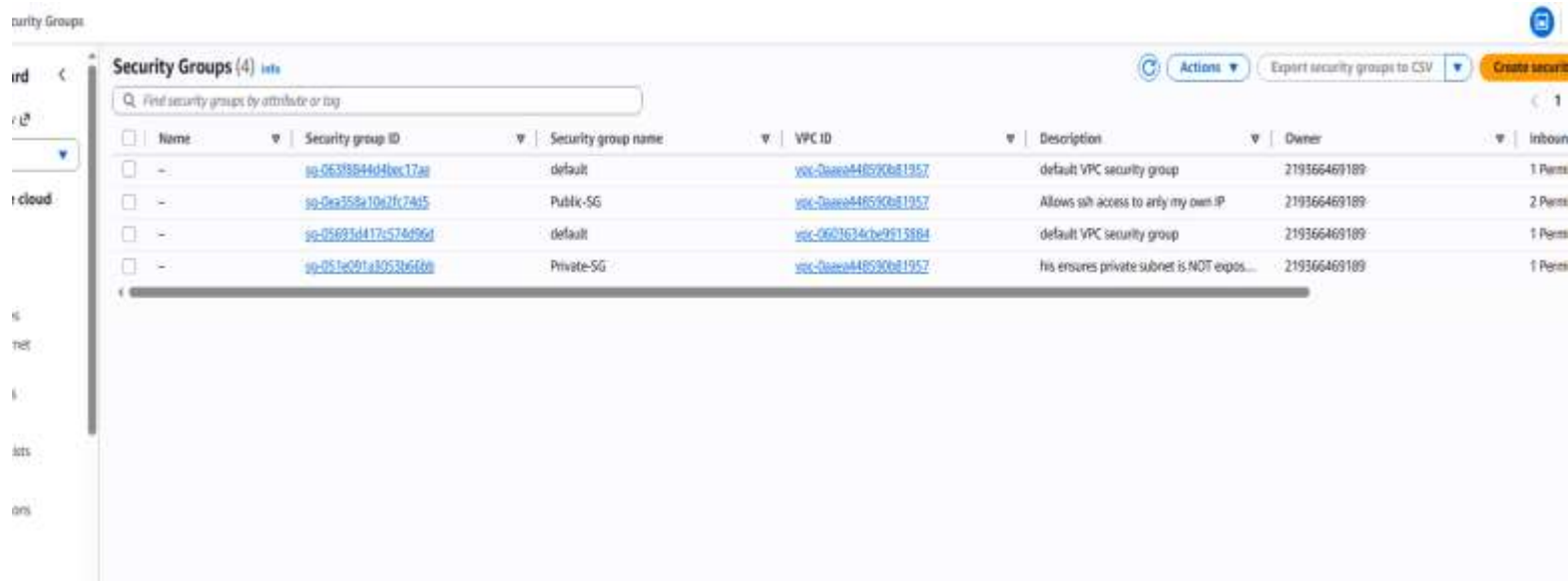
Private EC2 Security Group

Name: Private-SG

Rules:

SSH (22) → ONLY Public Subnet or Public SG

This ensures private subnet is NOT exposed.



Name	Security group ID	Security group name	VPC ID	Description	Owner	Inbound
default	sg-06338d4d4kx17a8	default	vpc-0a8a448590b81957	default VPC security group	219366469189	1 Permit
Public-SG	sg-0ea358a10a2f17415	Public-SG	vpc-0a8a448590b81957	Allows ssh access to only my own IP	219366469189	2 Permit
default	sg-05693d417c574d96d	default	vpc-0603634cbe9f13884	default VPC security group	219366469189	1 Permit
Private-SG	sg-051ac091a80531668	Private-SG	vpc-0a8a448590b81957	his ensures private subnet is NOT expos...	219366469189	1 Permit

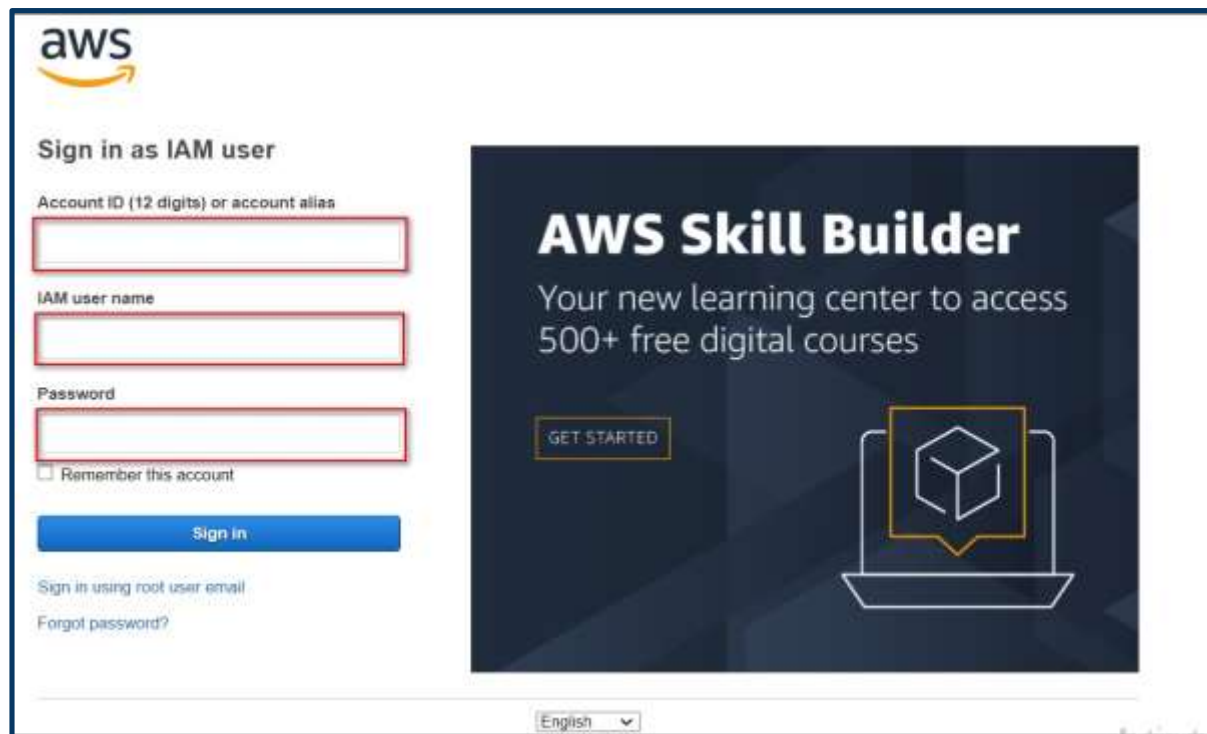
Figure 1.1.24: Private EC2 Groups

TASK 2. LAUNCHING AND CONFIGURING A LINUX EC2 INSTANCE AT CYBERARIANS

○ STEP 1: Launching an EC2 Instance in AWS

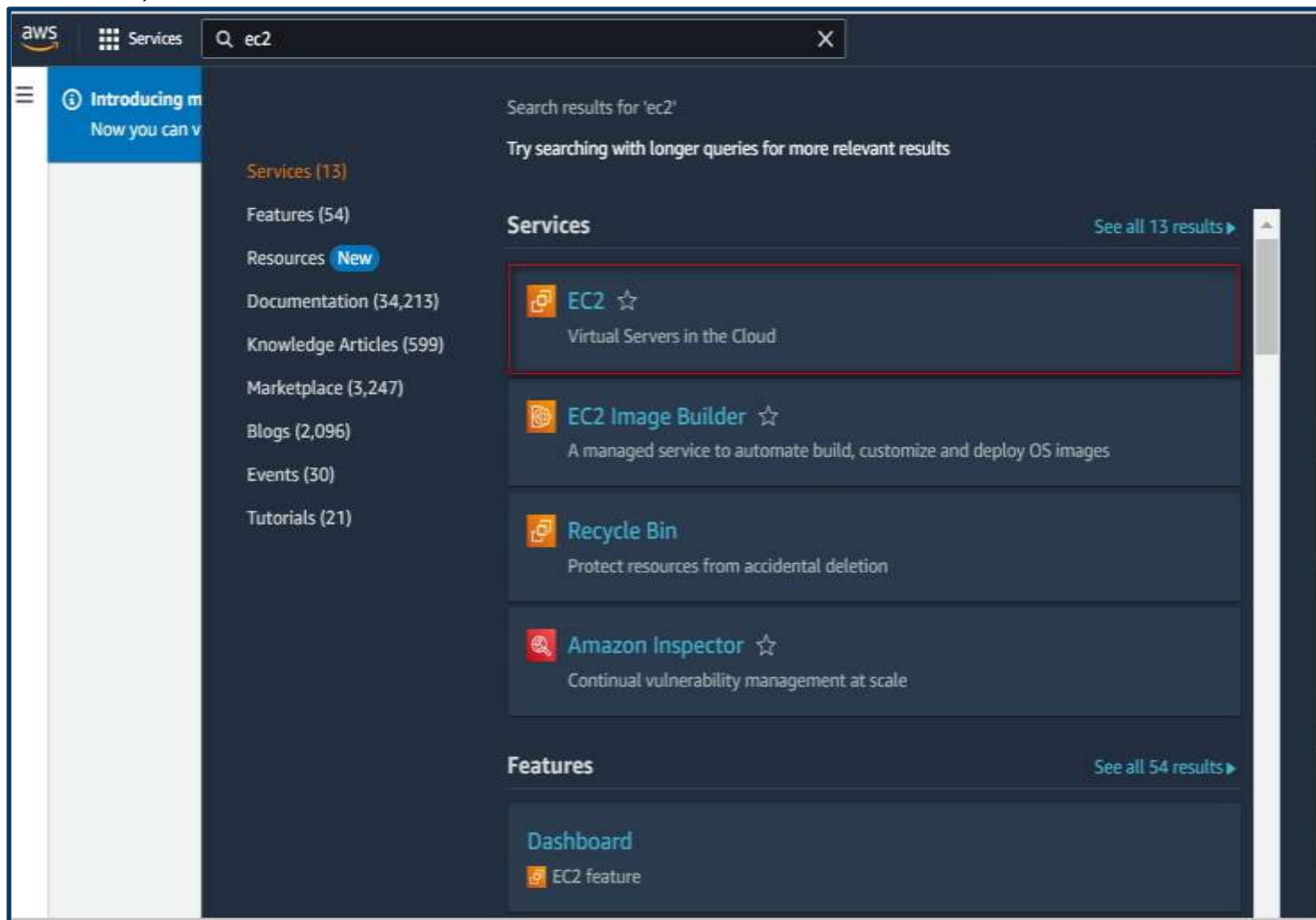
Sign in to AWS Console: Go to the AWS Management Console at

<https://console.aws.amazon.com/> (Figure 1.1: Search result).



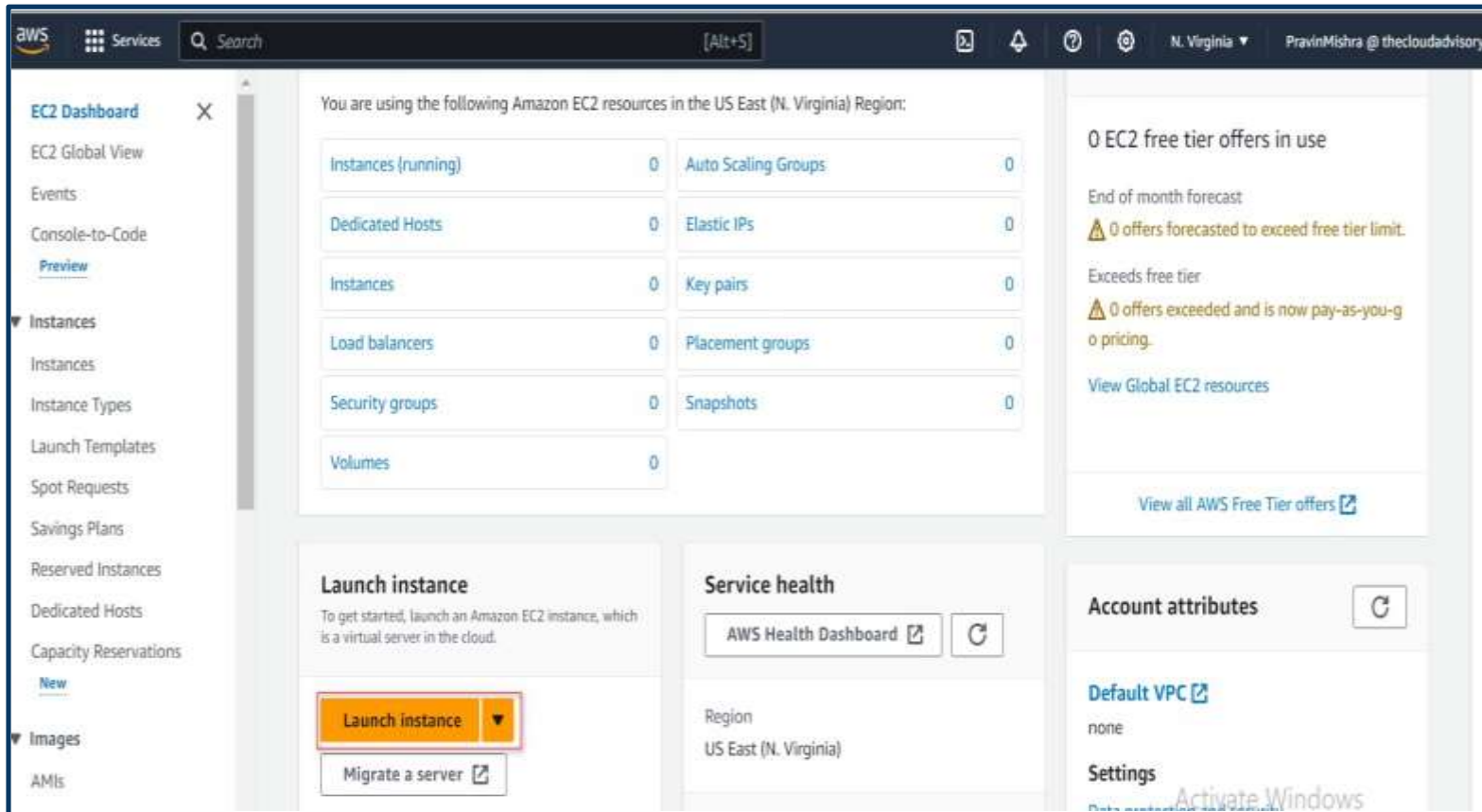
Navigate to EC2 Dashboard:

- Click on the "Services" dropdown in the top left corner.
- Select "EC2" under the "Compute" section. (*Figure 1.2: EC2 Search result*)



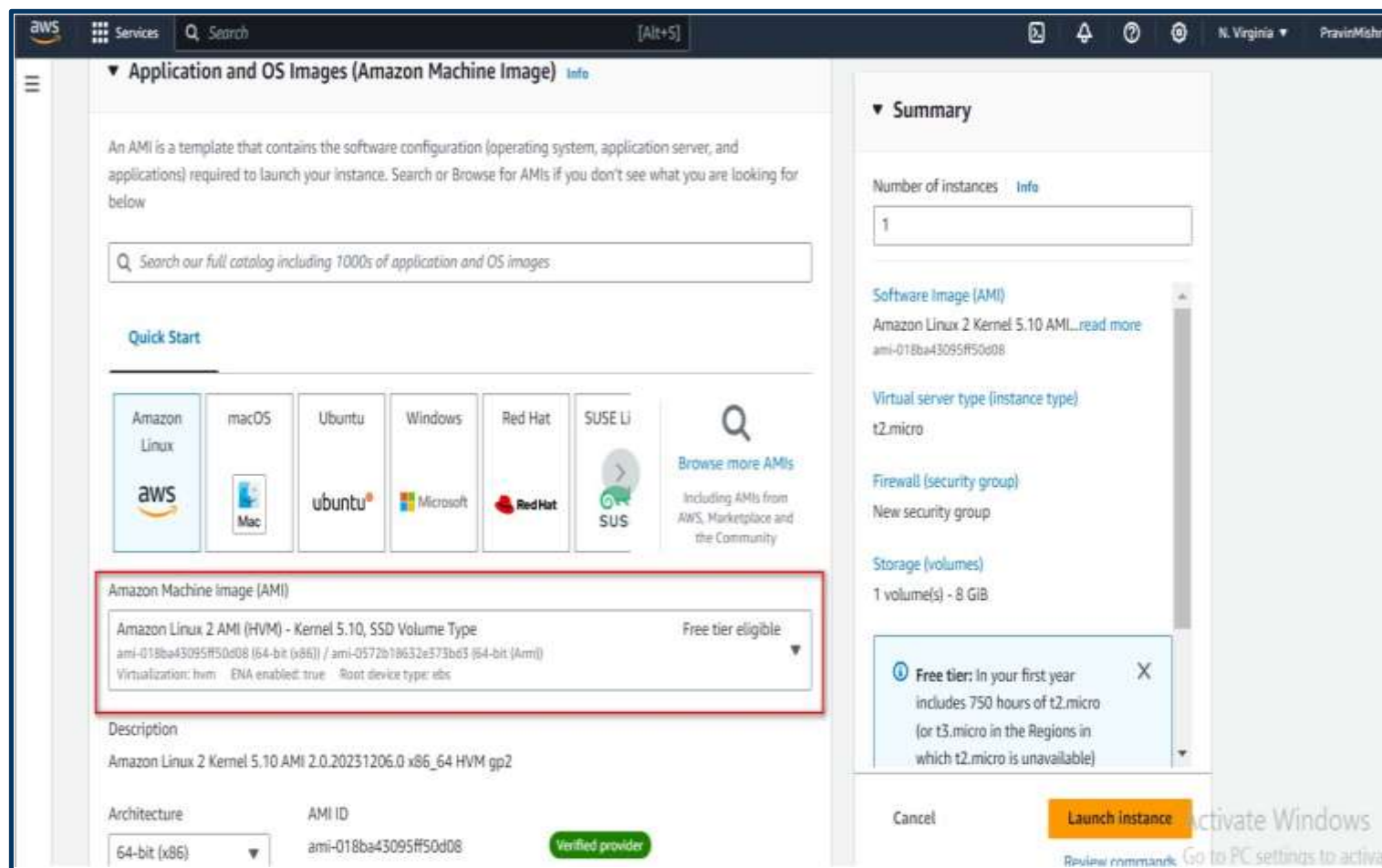
Launch Instance:

- Click on the "Instances" link in the EC2 Dashboard.
- Press the "Launch Instance" button. (Figure 1.3: Launch Instance Tab)



Choose an Amazon Machine Image (AMI):

- Select "Amazon Linux 2 AMI" from the list of available AMIs.
- Click on "Select" to proceed. *Figure 1.4: Choose an Amazon Machine Image (AMI)*



Choose t3.micro as the instance type for this exercise.

- (Figure 1.5: Choose an Instance Type)

The screenshot displays the AWS Management Console interface for creating a new EC2 instance. The 'Instance type' section is highlighted with a red box, showing the 't2.micro' instance type selected. The 'Summary' panel on the right provides a overview of the configuration, including the number of instances (1), the software image (Amazon Linux 2023 AMI), the virtual server type (t2.micro), the firewall (security group), and the storage (volumes). A 'Free tier' notification is also visible, indicating that the instance is eligible for the free tier.

Instance type Info | Get advice

Instance type

t2.micro Free tier eligible

Family: t2 1 vCPU 1 GiB Memory Current generation: true

On-Demand Windows base pricing: 0.0162 USD per Hour

On-Demand SUSE base pricing: 0.0116 USD per Hour

On-Demand RHEL base pricing: 0.0716 USD per Hour

On-Demand Linux base pricing: 0.0116 USD per Hour

Additional costs apply for AMIs with pre-installed software

Summary

Number of instances Info

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.3.2...read more

ami-0759f51a90924c166

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year includes 750 hours of t2.micro (or t3.micro in the Regions in which t2.micro is unavailable)

Cancel Launch instance

STEP 2: KEY PAIR CREATION

- Select “**Create a new key pair**” during EC2 setup.
- Enter a unique name for the key pair.
- Download the **.pem file** and store it securely.
- This key pair is required for secure SSH access to the Amazon EC2 instance.

Key pair name

Key pairs allow you to connect to your instance securely.

Cybrerians

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ **RSA**
 RSA encrypted private and public key pair

☐ **ED25519**
 ED25519 encrypted private and public key pair

Private key file format

☒ **.pem**
 For use with OpenSSH

☐ **.ppk**
 For use with PuTTY

⚠ When prompted, store the private key in a secure and accessible location on your computer. **You will need it later to connect to your instance.** [Learn more](#)

Cancel

Create key pair

Figure 1.6: Key Pair Creation

STEP 3: VPC AND NETWORK CONFIGURATION:

- Use an existing VPC, ensuring it's connected to a public subnet.
- Enable "Auto-assign public IP" to ensure the instance is accessible over the internet. (*Figure 1.7: VPC and Network Configuration*)

▼ Network settings

Info

VPC - required

Info

vpc-0aaea448590b81957 (Cyberarians-VPC)

10.0.0.0/16

↻

Subnet

Info

subnet-00b2ef23ae7b20cf4

1c

↻ Create new subnet

subnet-00b2ef23ae7b20cf4

VPC: vpc-0aaea448590b81957 Owner: 219366469189

Availability Zone: eu-north-1a (eun1-az1) Zone type: Availability Zone

IP addresses available: 249 CIDR: 10.0.3.0/24

Auto-assign public IP

Info

Disable

▼

Firewall (security groups)

Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group

☒ Select existing security group

Common security groups

Info

Select security groups

▼

↻ Compare security group rules

Security Group Cybrarians sg-0dbab47e3bb5bae3a

✕

VPC: vpc-0aaea448590b81957

Security groups that you add or remove here will be added to or removed from all your network interfaces.

STEP 4: SECURITY GROUP SETUP:

- Create a new security group for instance and specify the group name and description. (*Figure 1.8: Security Group Setup*)

Firewall (security groups) | [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group

☒ Select existing security group

Common security groups | [Info](#)

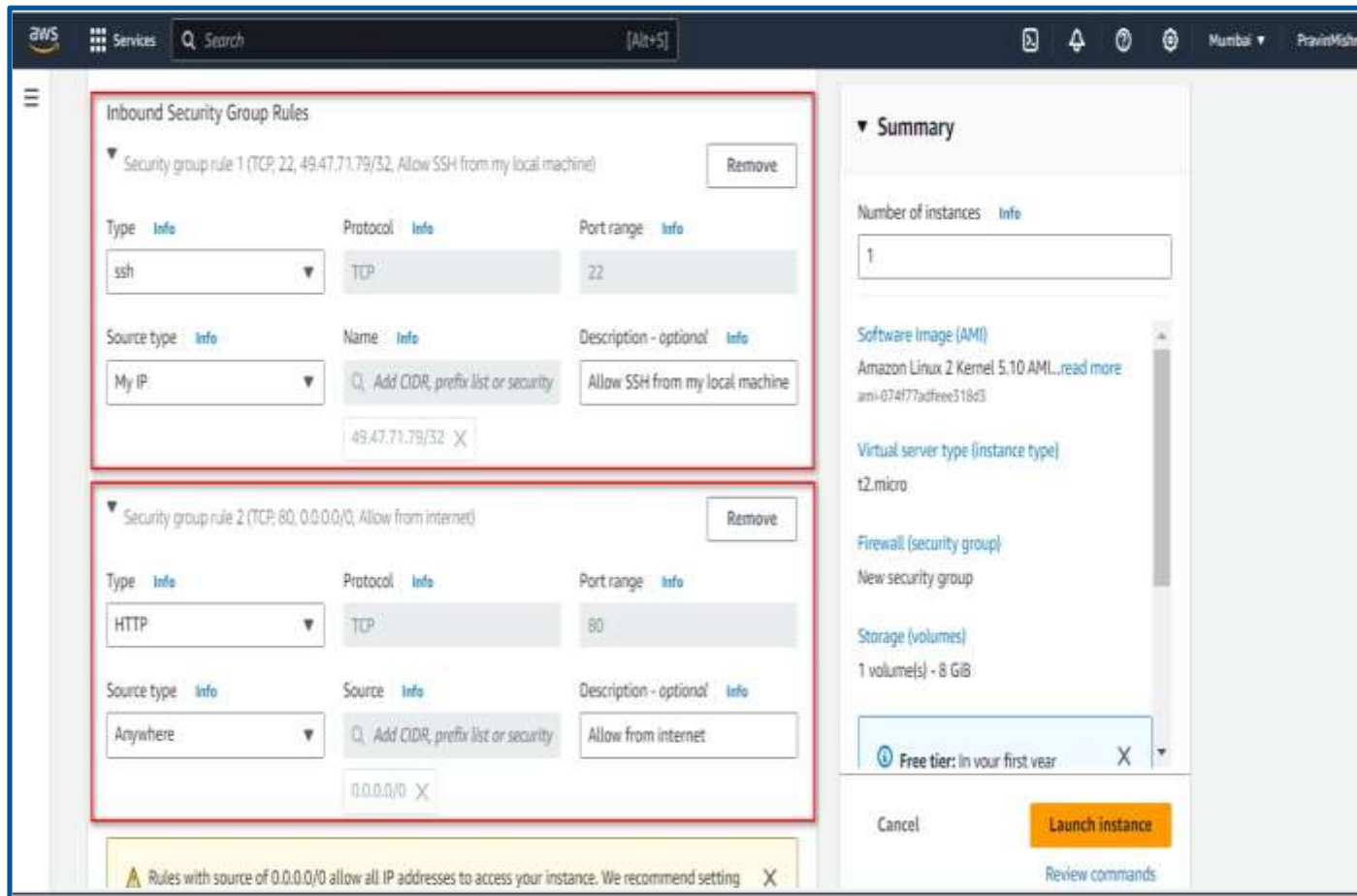
Select security groups

[Compare security group rules](#)

Security Group Cybrarians sg-0dbab47e3bb5bae3a [×](#)
VPC: vpc-0aaaa448590b81957

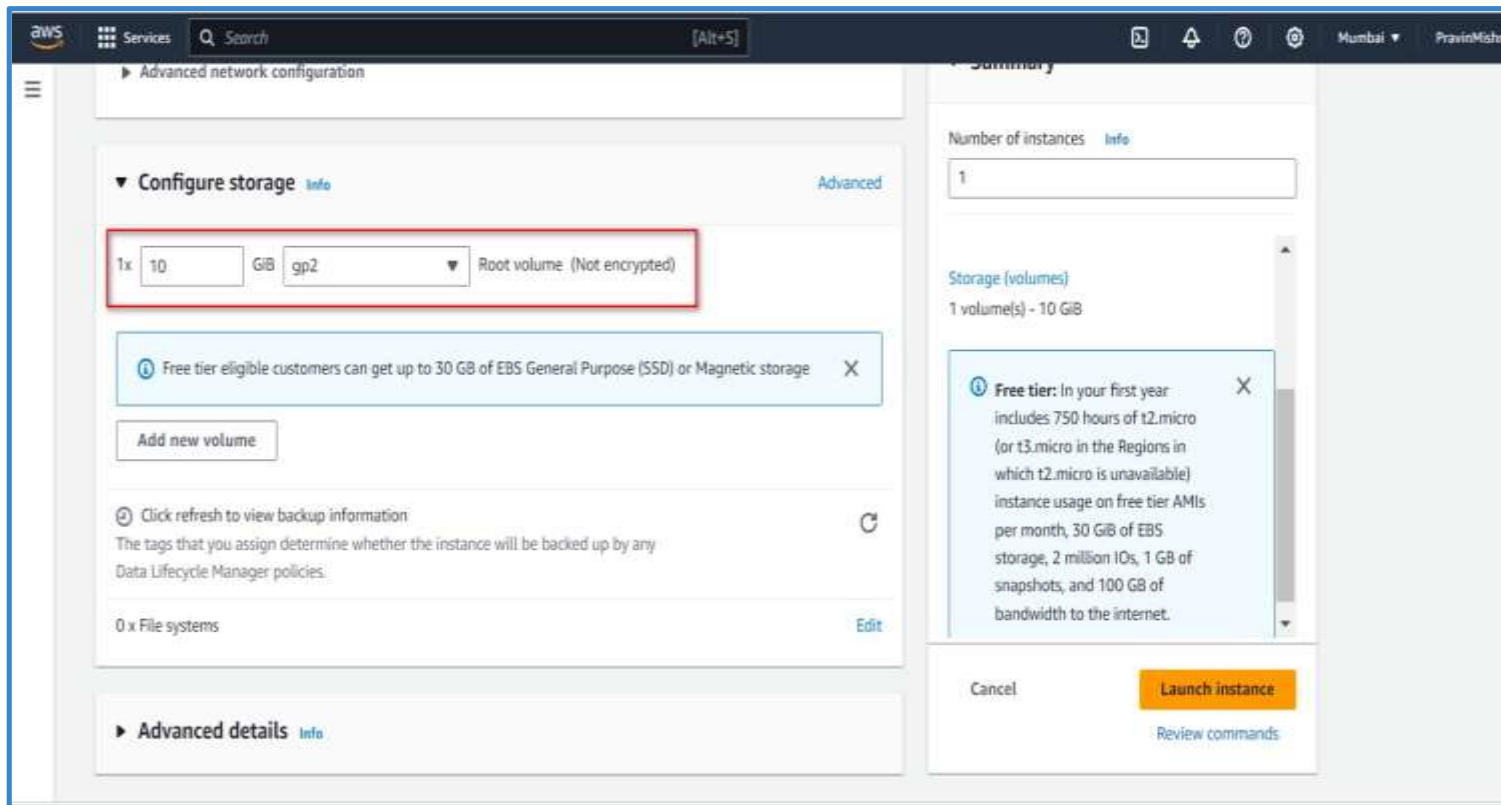
Security groups that you add or remove here will be added to or removed from all your network interfaces.

- Configure inbound rules to allow traffic on ports 22 (SSH) and 80 (HTTP) from the internet. (*Figure 1.9: Security Group Rules*).



STEP 5: STORAGE CONFIGURATION:

- Modify the instance's storage settings, setting the size to 10 GB. (*Figure 1.10: Storage Configuration*)



- Click “**Launch Instance**” to complete the setup and start your Amazon EC2 instance.

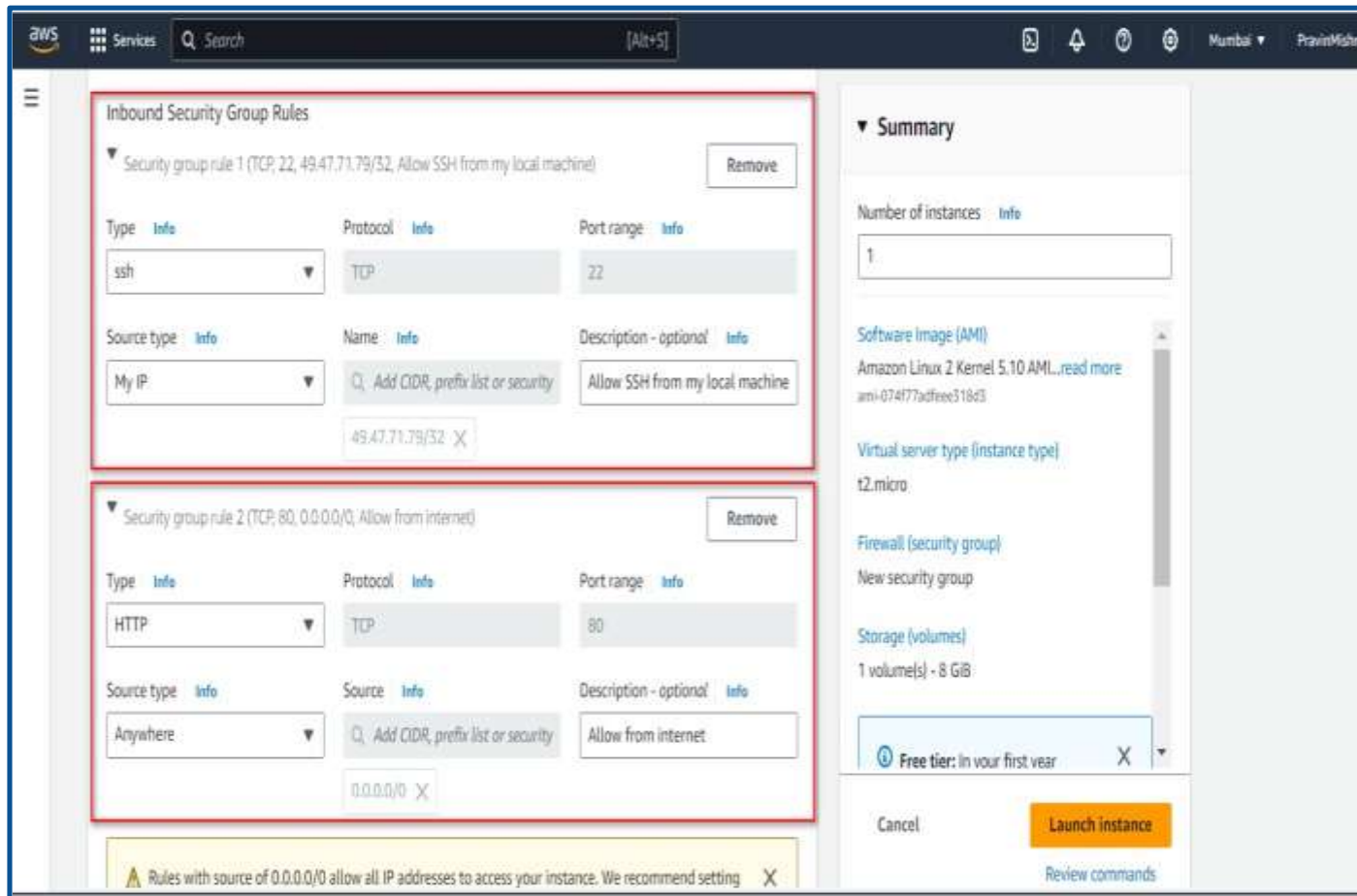


Figure 1.10: Launch Instance Final

STEP 6: CONNECTING TO EC2 VIA SSH

- a. Select your Amazon EC2 instance from the dashboard.
- b. Click “**Connect**” at the top.
- c. Open the **SSH client** tab.
- d. Copy the provided command (including username), e.g.:

```
ssh -i /path/to/your-key.pem ec2-user@your-instance-public-ipv4
```

EC2 Instance Connect	SSM Session Manager	<u>SSH client</u>	EC2 serial console
----------------------	---------------------	-------------------	--------------------

Instance ID
i-07e6d68617a021642 (Cybrarians Web Server)

1. Open an SSH client.
2. Locate your private key file. The key used to launch this instance is Cybrarians.pem
3. Run this command, if necessary, to ensure your key is not publicly viewable.
chmod 400 "Cybrarians.pem"
4. Connect to your instance using its Public IP:
16.171.31.59

Example:
ssh -i "Cybrarians.pem" ec2-user@16.171.31.59

Note: In most cases, the guessed username is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Figure 1.11: SSH Client Console

STEP 7: CONNECTING VIA COMMAND LINE (TERMINAL)

- a. Open a terminal (Linux/macOS) or use PuTTY / Git Bash on Windows.
 - b. Navigate to the folder containing your key pair:
`cd /path/to/your-key-pair-directory`
- c. Secure the key file permissions:
`chmod 400 your-key-pair.pem`
This ensures only the owner can read the file.
- d. Connect to your Amazon EC2 instance using SSH:
`ssh -i your-key-pair.pem ec2-user@your-instance-public-ipv4`
- Example:
`ssh -i "Keypair_Cyberarians.pem" ec2-user@13.232.87.112`
- e. After login, verify connectivity:
`ping google.com`
- **Note:** Incorrect file path or IP address is the most common cause of connection failure.

```
F223616@Girls-hostel MINGW64 ~/Downloads
$

F223616@Girls-hostel MINGW64 ~/Downloads
$ ssh -i "EC2-public.pem" ubuntu@13.61.144.128
The authenticity of host '13.61.144.128 (13.61.144.128)' can't be established.
ED25519 key fingerprint is: SHA256:biVv1YPnLPN8bwEZ1L+ZAu/b/nshDmI/+70cPuthetw
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '13.61.144.128' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1007-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Apr 16 19:30:47 UTC 2026

System load: 0.22           Temperature: -273.1 C
Usage of /: 26.3% of 6.71GB Processes: 111
Memory usage: 23%          Users logged in: 0
Swap usage: 0%             IPv4 address for ens5: 10.0.1.98

 * Ubuntu Pro delivers the most comprehensive open source security and
   compliance features.

https://ubuntu.com/aws/pro
```

Figure 1.12: Connecting via Command Line

TASK 3 : LAUNCHING, CONFIGURING, AND INSTALLING WORDPRESS ON A LINUX EC2 INSTANCE AT CYBERARIANS

Task Overview

After launching and connecting to your Linux Amazon EC2 instance, the next step is to install and configure WordPress. This simulates a real-world setup for Cyberarians to manage blogs, content, or an online book platform.

This task includes:

- Installing and configuring WordPress on an Amazon Linux 2 EC2 instance.
- Verifying that the website is publicly accessible and functioning correctly over the internet.

STEP 1: UPDATE AND UPGRADE PACKAGES

Once connected to your EC2 instance, update the package lists and upgrade installed packages by using `sudo yum update`: (*Figure 2.1: Update and Upgrade Packages*).

Command:

- *sudo yum update*

STEP 2: INSTALL APACHE WEB SERVER

- Install the Apache web server by using the below command (*Figure 2.2: Install Apache Web Server*).

Command:

- *sudo yum install httpd -y*

```
[ec2-user@ip-10-0-2-117 ~]$ sudo yum install httpd -y
Last metadata expiration check: 0:25:34 ago on Thu Apr 23 07:36:55 2026.
Package httpd-2.4.66-1.amzn2023.0.1.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-10-0-2-117 ~]$
```

Figure 2.2: Install Apache Web Server

Start the Apache service and enable it to start on boot (*Figure 2.3: Apache service status*)

Command:

- *sudo systemctl start httpd*
- *sudo systemctl status httpd*

```
[ec2-user@ip-10-0-2-117 ~]$ sudo systemctl start httpd
[ec2-user@ip-10-0-2-117 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: d>
   Drop-In: /usr/lib/systemd/system/httpd.service.d
            └─php-fpm.conf
```

Figure 2.3: Apache service status

- To verify the deployment of Apache HTTPD through the browser:
- Obtain the public IP address of your EC2 instance (*Figure 2.4: Public IP address*).
- Open your web browser.
- Enter the EC2 instance's public IP address into the browser's address bar.
- Press "Enter" to load the page (*Figure 2.5: Apache Test Page*).
- Example: `http://[EC2_Public_IP]`



Figure 2.4: Public IP address



Figure 2.5: Apache Test Page

STEP 3: INSTALL PHP AND MYSQL

Install PHP along with necessary PHP extensions required by WordPress(*Figure 2.6: install PHP*).

Command:

- *sudo amazon-linux-extras install -y lamp-mariadb10.2-php7.2 php7.2*

```
ec2-user@ip-10-0-2-117:~  
[ec2-user@ip-10-0-2-117 ~]$ sudo dnf install -y httpd php php-mysqlnd mariadb105  
Last metadata expiration check: 16:10:29 ago on Sat Apr 18 13:54:30 2026.  
Package httpd-2.4.66-1.amzn2023.0.1.x86_64 is already installed.  
Dependencies resolved.  
=====
```

Package	Architecture	Vers
Installing:		
mariadb105	x86_64	3:10
php8.5	x86_64	8.5.
php8.5-mysqlnd	x86_64	8.5.
Installing dependencies:		

Figure 2.6: install PHP

Command:

- *sudo yum install -y mysql*

```
ec2-user@ip-10-0-2-117:~  
[ec2-user@ip-10-0-2-117 ~]$ sudo yum install -y mysql
```

Figure 2.7: install MySQL

Command:

- *sudo yum install -y mysql*

```
ec2-user@ip-10-0-2-117:~  
[ec2-user@ip-10-0-2-117 ~]$ sudo yum install -y mysql  
Last metadata expiration check: 16:11:45 ago on Sat Apr 18 13:54:30 2026.  
No match for argument: mysql  
Error: Unable to find a match: mysql  
[ec2-user@ip-10-0-2-117 ~]$ sudo dnf install -y mariadb105  
Last metadata expiration check: 16:12:46 ago on Sat Apr 18 13:54:30 2026.  
Package mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64 is already installed.  
Dependencies resolved.  
Nothing to do.  
Complete!  
[ec2-user@ip-10-0-2-117 ~]$ mysql --version  
mysql Ver 15.1 Distrib 10.5.29-MariaDB, for Linux (x86_64) using EditLine wrapper  
[ec2-user@ip-10-0-2-117 ~]$ |
```

Figure 2.7: install MySQL

STEP 4: WORDPRESS SETUP

- Download and install WordPress on the EC2 instance(*Figure 2.8: Download WordPress*)

Navigate to the Web Directory: Move to the directory where you want to install WordPress. For instance, in

- `cd /var/www/html/`

Download WordPress: Fetch the latest version of WordPress.

Command:

- `sudo wget https://wordpress.org/latest.tar.gz`

Extract WordPress Archive: Unzip the downloaded WordPress file.

Command:

- `sudo tar -xzf latest.tar.gz`

```
[ec2-user@ip-10-0-2-117 ~]$ cd /var/www/html/  
[ec2-user@ip-10-0-2-117 html]$ pwd  
/var/www/html  
[ec2-user@ip-10-0-2-117 html]$ sudo wget https://wordpress.org/latest.tar.gz  
--2026-04-19 06:09:21-- https://wordpress.org/latest.tar.gz  
Resolving wordpress.org (wordpress.org)... 198.143.164.252, 2607:f978:5:8002::c68f:a4fc  
Connecting to wordpress.org (wordpress.org)|198.143.164.252|:443... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 27064862 (26M) [application/octet-stream]  
Saving to: 'latest.tar.gz'
```

```
latest.tar.gz 100%[=====]
```

```
2026-04-19 06:09:24 (13.9 MB/s) - 'latest.tar.gz' saved [27064862/27064862]
```

```
[ec2-user@ip-10-0-2-117 html]$
```

```
[ec2-user@ip-10-0-2-117 html]$ sudo tar -xzf latest.tar.gz
```

```
wordpress/
```

```
wordpress/index.php
```

```
wordpress/license.txt
```

```
wordpress/readme.html
```

```
wordpress/wp-activate.php
```

```
wordpress/wp-admin/
```

```
wordpress/wp-admin/about.php
```

```
wordpress/wp-admin/admin-ajax.php
```

```
wordpress/wp-admin/admin-footer.php
```

```
wordpress/wp-admin/admin-functions.php
```

```
wordpress/wp-admin/admin-header.php
```

```
wordpress/wp-admin/admin-post.php
```

Command:

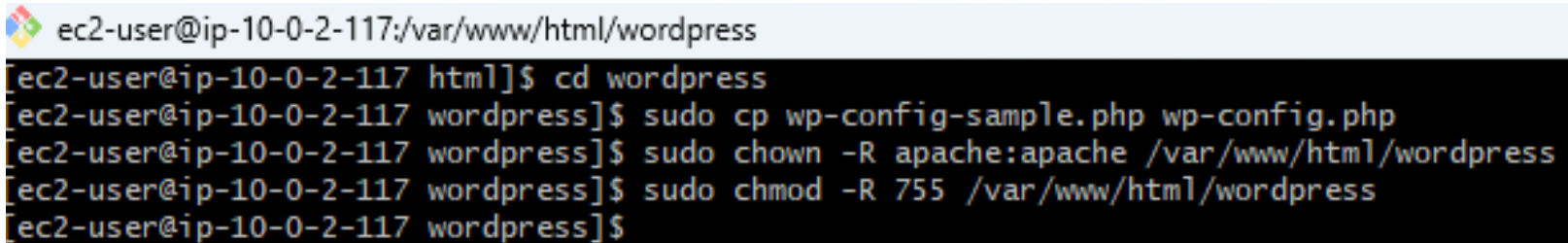
- *cd wordpress*

Create the configuration file by copying the sample configuration.

sudo cp wp-config-sample.php wp-config.php

Set Permissions: Adjust the permissions for WordPress files.

- *sudo chown -R apache:apache /var/www/html/wordpress*
sudo chmod -R 755 /var/www/html/wordpress



```
ec2-user@ip-10-0-2-117:/var/www/html/wordpress
ec2-user@ip-10-0-2-117 html]$ cd wordpress
ec2-user@ip-10-0-2-117 wordpress]$ sudo cp wp-config-sample.php wp-config.php
ec2-user@ip-10-0-2-117 wordpress]$ sudo chown -R apache:apache /var/www/html/wordpress
ec2-user@ip-10-0-2-117 wordpress]$ sudo chmod -R 755 /var/www/html/wordpress
ec2-user@ip-10-0-2-117 wordpress]$
```

Figure 2.9: Permission for Wordpress file

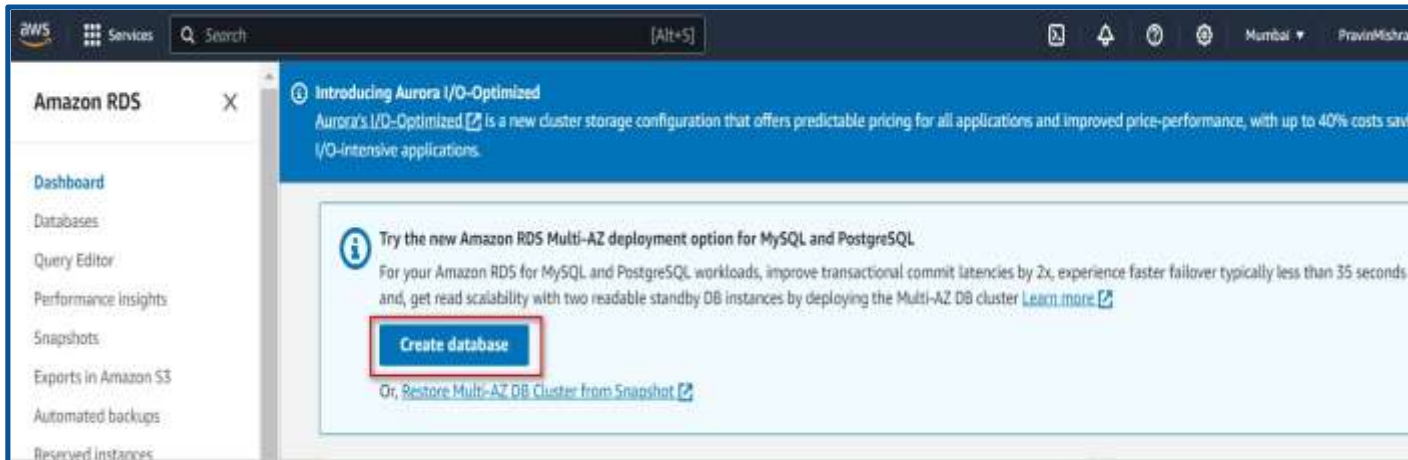
STEP 5: CREATE WORDPRESS DATABASE

Before even creating the Database, creating a subnet group is important

Navigate to the left navigation pane of the **RDS dashboard** and click on the *Subnet groups* (*Figure 2.10: Create Subnet groups*)

Figure 2.10: Create Subnet groups

Navigate to the RDS console and click on "Create Database." (*Figure 2.10: Create Database Tab*)



- Under Engine options, choose "MySQL." (*Figure 2.11: Create Database Option*)

Create database [Info](#)

Engine options

Engine type [Info](#)

☐ Aurora (MySQL Compatible)


☐ Aurora (PostgreSQL Compatible)


☒ MySQL


☐ PostgreSQL


☐ MariaDB


☐ Oracle


☐ Microsoft SQL Server


☐ IBM Db2


Choose a database creation method

☒ Full configuration
You set all of the configuration options, including ones for availability, security, backups, and maintenance.

☐ Easy create
Use the recommended best-practice configurations. Some configuration options can be changed after the database is created.

Templates

Choose a sample template to meet your use case.

☐ Production
Use defaults for high availability and fast, consistent performance.

☒ Dev/Test
This instance is intended for development use outside of a production environment.

☐ Sandbox
To develop new applications, test existing applications, or gain hands-on experience with Amazon RDS.

Availability and durability

Deployment options [Info](#)

Choose the deployment option that provides the availability and durability needed for your use case. RDS is committed to a certain level of uptime depending on the deployment option you choose. Learn more in the [Amazon RDS service level agreement \(SLA\)](#).

☐ Multi-AZ DB cluster deployment (3 instances)
Creates a primary DB instance with two readable standbys in separate Availability Zones. This setup provides:

- 99.95% uptime
- Redundancy across Availability Zones
- Increased read capacity
- Reduced write latency

Write/read endpoint
AZ 1

Reader endpoints
AZ 2
Readable standby + SSD

☐ Multi-AZ DB instance deployment (2 instances)
Creates a primary DB instance with a non-readable standby instance in a separate Availability Zone. This setup provides:

- 99.95% uptime
- Redundancy across Availability Zones

Write/read endpoint
AZ 1

Standby (no endpoint)
AZ 2

☒ Single-AZ DB instance deployment (1 instance)
Creates a single DB instance without standby instances. This setup provides:

- 99.9% uptime
- No data redundancy

Write/read endpoint
AZ 1

© 2026, Amazon Web Services

In the Settings section(*Figure 2.13:Credentials Settings*):
Enter “admin” as the DB identifier.
Set your desired Master username.
Define a secure Master password (e.g. Happynewyear#24) for access
and also enter the same password on the confirm password tab.

Engine version

MySQL 8.4.8

☐ Enable RDS Extended Support [Info](#)
Amazon RDS Extended Support is a [paid offering](#). By selecting this option, you consent to being charged for this offering if you are running your database major version past the RDS end of standard support date for the major version in the [RDS for MySQL documentation](#).

DB instance identifier [Info](#)
Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.

Cyberarians-db

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens.

Credentials Settings

Master username [Info](#)
Type a login ID for the master user of your DB instance.

admin

1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management
You can use AWS Secrets Manager or manage your master user credentials.

☐ Managed in AWS Secrets Manager - *most secure*
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☒ Self managed
Create your own password or have RDS create a password that you manage.

☐ Auto generate password
Amazon RDS can generate a password for you, or you can specify your own password.

Master password [Info](#)

Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / * @

The instance configuration and storage should remain the same at their default selections(*Figure 2.14:Instance Configuration*).

Use the default configuration for **Burstable Classes** with the **db.t3.micro** instance type.

Specify the storage type as **General Purpose SSD (gp2)**.

Allocate **20 GB** of storage capacity.

The screenshot displays the AWS Management Console interface for configuring an Amazon RDS instance. The top navigation bar includes the AWS logo, a 'Services' menu, a search bar, and a keyboard shortcut '[Alt+S]'. The main content area is titled 'Instance configuration' and includes a note: 'The DB instance configuration options below are limited to those supported by the engine that you selected above.'

Under the 'DB instance class' section, there is a 'Hide filters' button and three toggle switches: 'Show instance classes that support Amazon RDS Optimized Writes' (disabled), 'Include previous generation classes' (disabled), and 'Standard classes (includes m classes)' (disabled). Below these are three radio button options: 'Standard classes (includes m classes)', 'Memory optimized classes (includes r and x classes)', and 'Burstable classes (includes t classes)' (selected). A dropdown menu shows 'db.t3.micro' with details '2 vCPUs 1 GiB RAM Network: 2,085 Mbps'.

The 'Storage' section features a 'Storage type' dropdown menu set to 'General Purpose SSD (gp2)' with the note 'Baseline performance determined by volume size'. Below this is an 'Allocated storage' input field set to '20' with a unit of 'GiB' and a note: 'The minimum value is 20 GiB and the maximum value is 6,144 GiB'.

In the Connectivity section(*Figure 2.15:Compute resource Configuration*): Choose the VPC named "VPC-Cyberarians" created in the preceding step.

For "Public access," select "No" to limit public access to the instance.

Connectivity [Info](#)

Compute resource

Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

☒ Don't connect to an EC2 compute resource

Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☐ Connect to an EC2 compute resource

Set up a connection to an EC2 compute resource for this database.

Virtual private cloud (VPC) [Info](#)

Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

Cyberarians-VPC (vpc-0aaea448590b81957)

4 Subnets, 2 Availability Zones



After a database is created, you can't change its VPC. Only VPCs with a corresponding DB subnet group are listed.

① After a database is created, you can't change its VPC.

DB subnet group [Info](#)

Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

default-vpc-0aaea448590b81957

4 Subnets, 2 Availability Zones

Public access [Info](#)

☐ Yes

RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.

☒ No

RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

Under "VPC security group," choose "Create new" and input "db-sg" as the name for the new VPC security group (*Figure 2.16: VPC Security Group*).

VPC security group (firewall) [Info](#)

Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

☐ Choose existing
Choose existing VPC security groups

☒ Create new
Create new VPC security group

New VPC security group name

RDS-SG

Availability Zone [Info](#)

eu-north-1a

RDS Proxy

RDS Proxy is a fully managed, highly available database proxy that improves application scalability, resiliency, and security.

☐ Create an RDS Proxy [Info](#)

RDS automatically creates an IAM role and a Secrets Manager secret for the proxy. RDS Proxy has additional costs. For more information, see [Amazon RDS Proxy pricing](#).

Certificate authority - optional [Info](#)

Using a server certificate provides an extra layer of security by validating that the connection is being made to an Amazon database. It does so by checking the server certificate that is automatically installed on all databases that you provision.

rdc-ca-rsa2048-g1 (default)

Expiry: May 25, 2061

If you don't select a certificate authority, RDS chooses one for you.

► Additional configuration

STEP 6: SECURITY GROUPS SETTINGS

- **Configuring Security Group Rules**
- Go to **Security Groups** and select “**public-instance-sg.**”
 - a. Click “**Edit inbound rules.**”
 - b. Add **MySQL/Aurora** rule:
 - Click “**Add rule**”
 - Type: **MySQL/Aurora**
 - Source: **Custom** → select “**db-sg**”
 - Save
 - c. Add **HTTP** rule:
 - Click “**Add rule**”
 - Type: **HTTP**
 - Source: **My IP**
 - Save
- These rules allow database communication and web access for your Amazon EC2 setup.

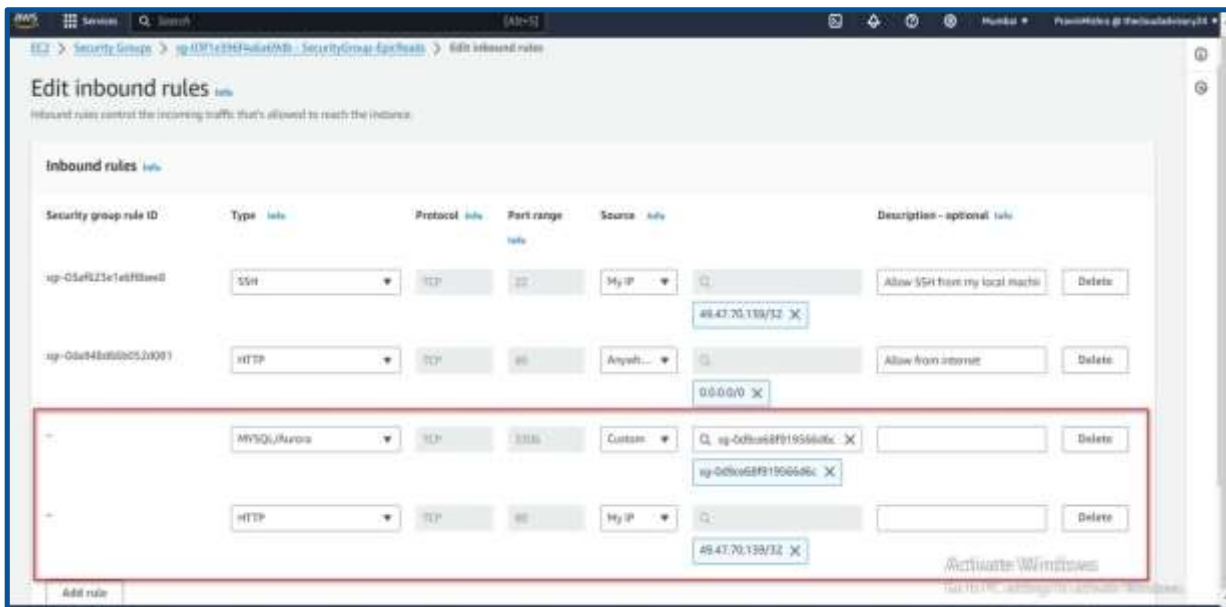


Figure 2.20: Security Group Cyberarians

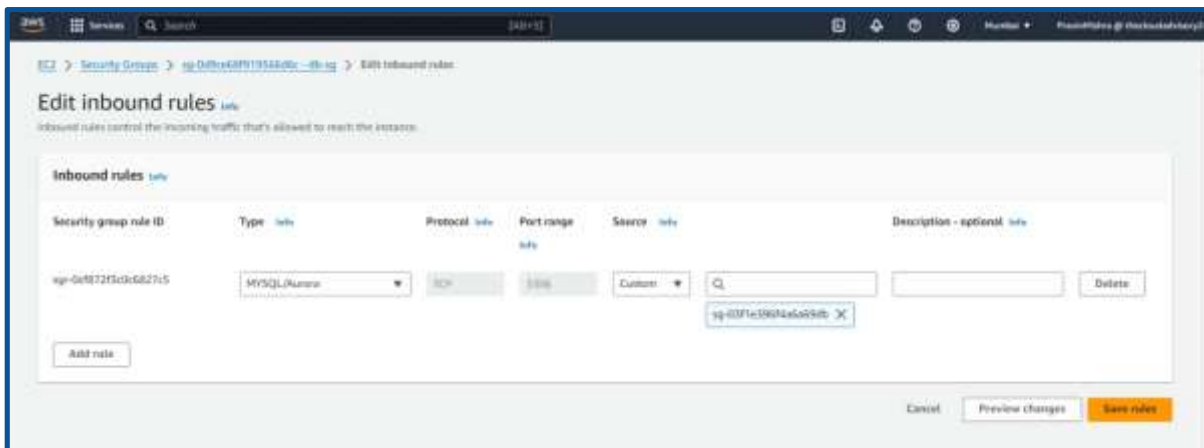


Figure 2.21: Security Group DB

STEP 7: CONFIGURE THE WORDPRESS INSTALLATION AND CONNECT IT TO A DATABASE

- Set MySQL environment variable:
`export MYSQL_HOST=<your-endpoint>`
This sets your Amazon RDS endpoint for database connectivity.
- Connect to WordPress database:
`mysql --user=<your-username> --password=<your-password> wordpress`
This logs into the database using MySQL credentials.
- Create user and grant permissions:
- `CREATE USER 'wordpress'@'localhost' IDENTIFIED BY 'wordpress-pass';`
`GRANT ALL PRIVILEGES ON wordpress.* TO wordpress;`
`FLUSH PRIVILEGES;`
- This creates a WordPress database user and assigns required privileges.
- Exit MySQL:
`EXIT;`
- These steps link Amazon RDS with your WordPress installation for proper backend functionality.

```
[ec2-user@ip-10-0-2-11/ wordpress]$ mysql -h $MYSQL_HOST -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 791
Server version: 8.4.8 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
4 rows in set (0.016 sec)

MySQL [(none)]> CREATE DATABASE wordpress;
Query OK, 1 row affected (0.020 sec)

MySQL [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
| wordpress |
+-----+
5 rows in set (0.002 sec)

MySQL [(none)]>
```

Figure 2.22: DB Connection page

STEP 7: CONFIGURE THE WORDPRESS INSTALLATION AND CONNECT IT TO A DATABASE

- Locate and Modify the Database Settings(*Figure 2.24: Nano text editor page*):
- Inside the wp-config.php file, find the section with MySQL settings. Replace the following placeholders:
 - `database_name_here` with the actual name of your WordPress database ('wordpress' in your case).
 - `username_here` with the MySQL database username ('wordpress' in your case).
 - `password_here` with the MySQL database password ('wordpress-pass' in your case).

```

<?php
/**
 * The base configuration for WordPress
 *
 * The wp-config.php creation script uses this file during the installation.
 * You don't have to use the web site, you can copy this file to "wp-config.php"
 * and fill in the values.
 *
 * This file contains the following configurations:
 *
 * * Database settings
 * * Secret keys
 * * Database table prefix
 * * ABSPATH
 *
 * @link https://wordpress.org/documentation/article/editing-wp-config-php/
 *
 * @package WordPress
 */

/** Database settings - You can get this info from your web host ** //
/** The name of the database for WordPress */
define( 'DB_NAME', 'wordpress' );

/** Database username */
define( 'DB_USER', 'admin' );

/** Database password */
define( 'DB_PASSWORD', 'Happynewyear#24' );

/** Database hostname */
define( 'DB_HOST', 'wordpress.cvkowic2mt4t.ap-south-1.rds.amazonaws.com' );

/** Database charset to use in creating database tables. */
define( 'DB_CHARSET', 'utf8' );

/** The database collate type. Don't change this if in doubt. */
define( 'DB_COLLATE', '' );

/**#@+
 * Authentication unique keys and salts.
 *
 * Change these to different unique phrases! You can generate these using
 * the {@link https://api.wordpress.org/secret-key/1.1/salt/ WordPress.org secret-key service}.
 */

```

^G Get Help ^O Write Out ^W Where Is ^K Cut Text ^J Justify ^C Cur Pos M-U Undo
 ^X Exit ^R Read File ^\ Replace ^U Uncut Text ^T To Spell ^_ Go To Line M-E Redo

Figure 2.24: Nano text editor page

- Run commands to deploy Wordpress on your computer

```
sudo yum install php-xml -y cd ..sudo cp -r  
wordpress/* /var/www/html  
sudo chown -R apache:apache /var/www/html Finally, start  
hosting the Apache web server sudo systemctl start  
httpd.service  
sudo systemctl enable httpd.service sudo systemctl  
restart php-fpm
```

Install WordPress through the Browser:

Open a web browser and enter your EC2 instance's **public IP address/wp-admin** or domain name(*Figure 2.27: Wordpress language page*).

```
http://{ec2-instance-public-IP-address}/wp-admin/
```

a. Follow the WordPress setup wizard:

Select the language and provide database connection details (database name, username, password, and host).

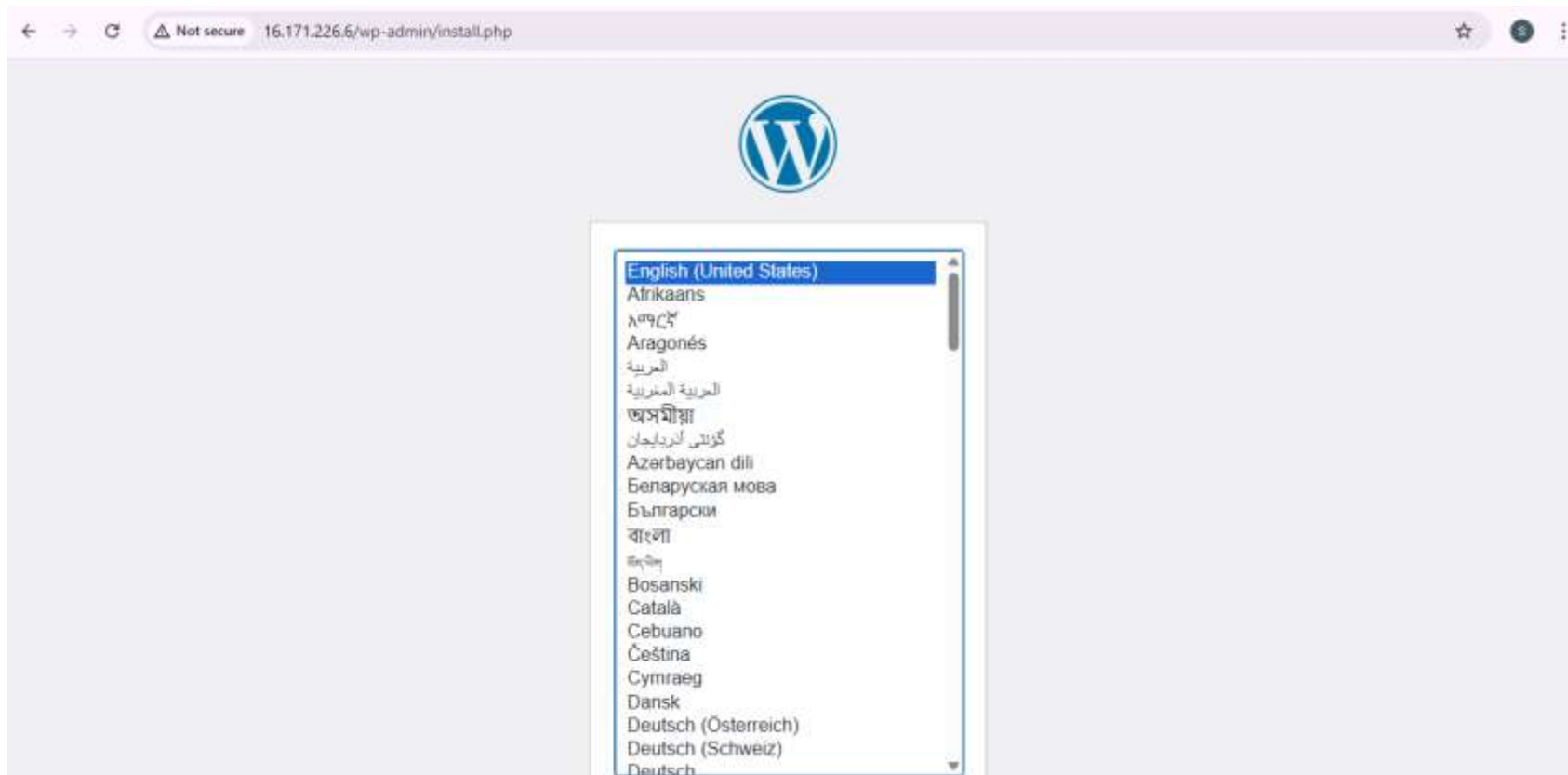


Figure 2.27: Wordpress language page

b. Complete the WordPress setup by providing site details (site title, admin username, password, email, etc.)(*Figure 2.28: Wordpress site details*).

Information needed

Please provide the following information. Do not worry, you can always change these settings later.

Site Title

WordPress

Username

Admin

Usernames can have only alphanumeric characters, spaces, underscores, hyphens, periods, and the @ symbol.

Password

R^r4@ktHG#xG10SiSD

Hide

Strong

Important: You will need this password to log in. Please store it in a secure location.

Your Email

sabahat.jahangir.tech@gmail.com

Double-check your email address before continuing.

Search engine visibility

☐ Discourage search engines from indexing this site

It is up to search engines to honor this request.

Install WordPress

Figure 2.28: Wordpress site details

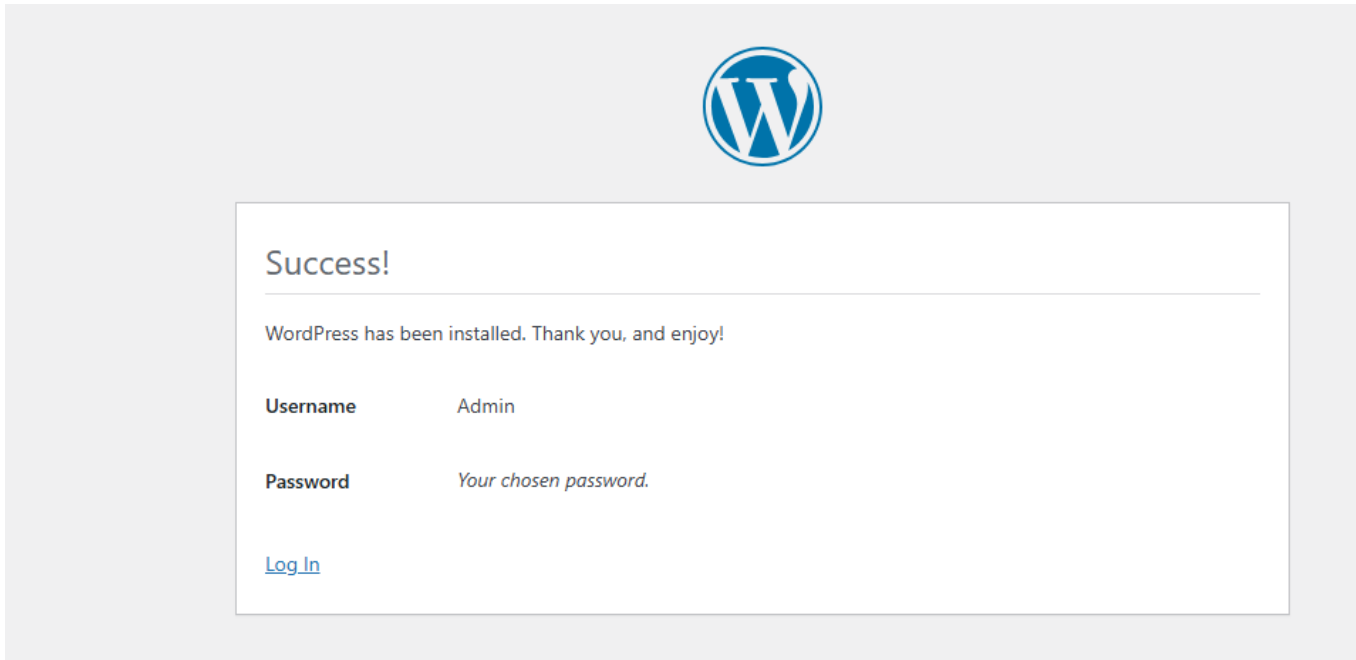


Figure 2.28: Wordpress site details

Login to the WordPress admin dashboard(*Figure 2.30:*
WordPress admin dashboard)

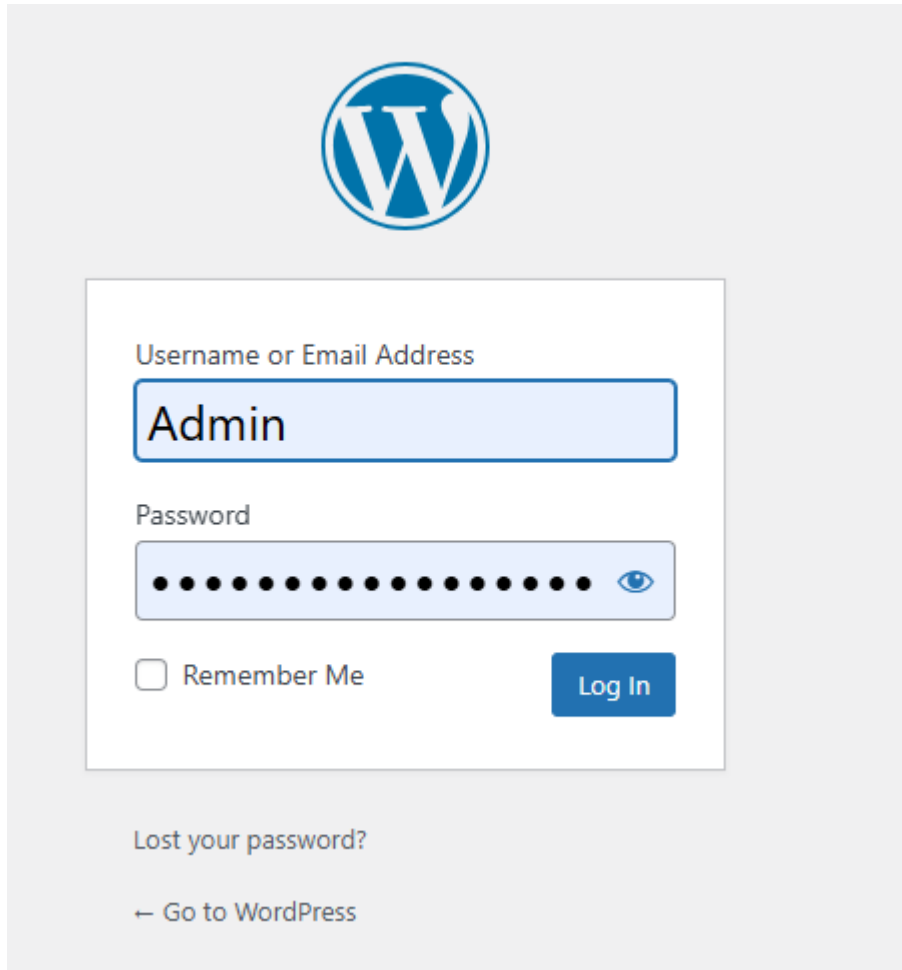
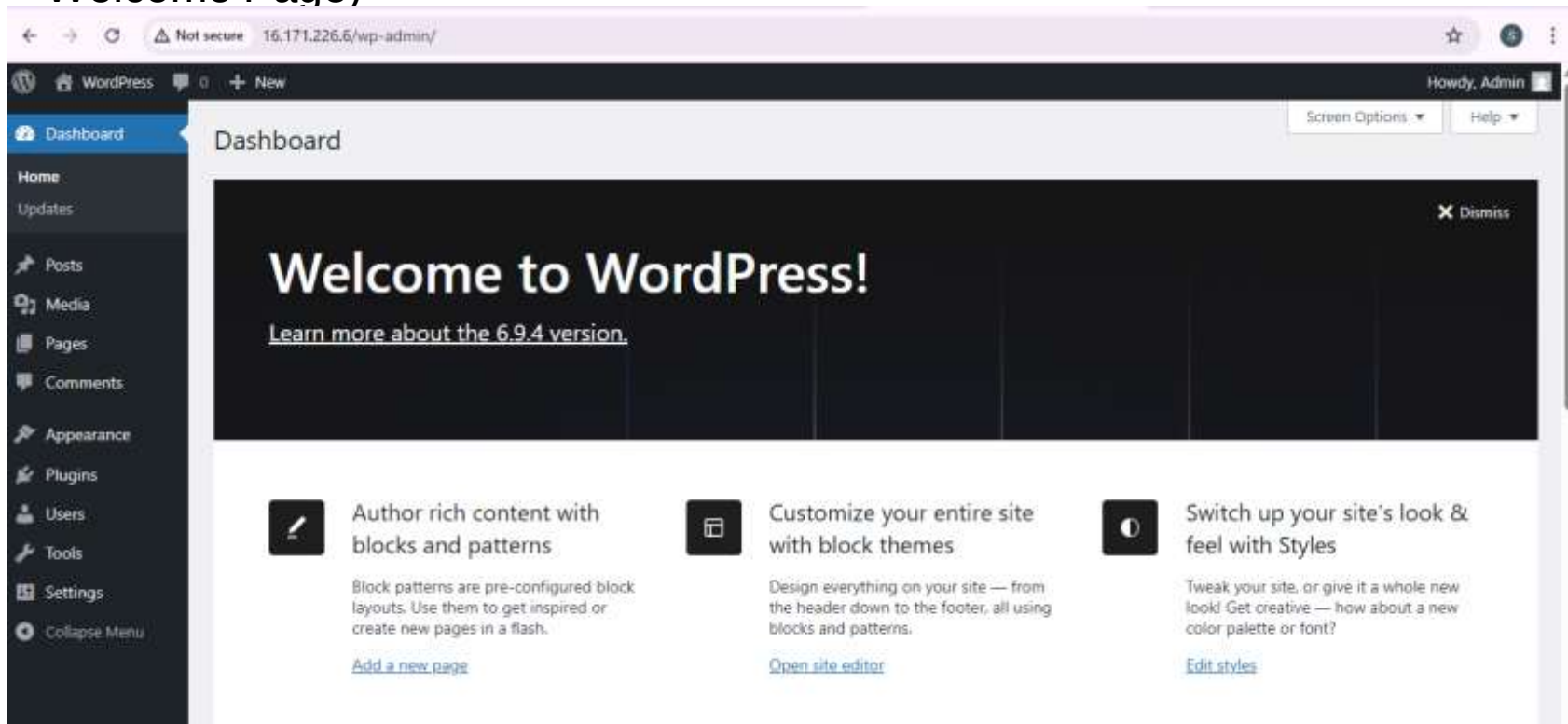
The image shows the WordPress login interface. At the top center is the WordPress logo, a blue circle with a white 'W'. Below it is a white rectangular box containing the login fields. The first field is labeled 'Username or Email Address' and contains the text 'Admin'. The second field is labeled 'Password' and contains a series of black dots, with an eye icon to its right for toggling visibility. Below the password field is a checkbox labeled 'Remember Me'. To the right of the checkbox is a blue button with the text 'Log In'. Below the login box, there is a link 'Lost your password?' and a link '← Go to WordPress'.

Figure 2.31: WordPress admin dashboard

"Great job! You're almost there! This is the final step where you'll access the WordPress app's main page. You've successfully launched, set up, and installed WordPress on your Linux EC2 Instance at Cyberarians. Let's take the last exciting step together!" (*Figure 2.32: WordPress Welcome Page*)



4/26/2026

AWS Cloud Security and Infrastructure
Implementation

Figure 2.32: WordPress Welcome Page

Ensure that the WordPress installation is functioning correctly by creating a test post or page(*Figure 2.33: Add New Post Page*). Steps to Create a Test Post/Page:

Navigate to Posts or Pages: In the WordPress Dashboard sidebar, click on "Add New Post" to create a test post.

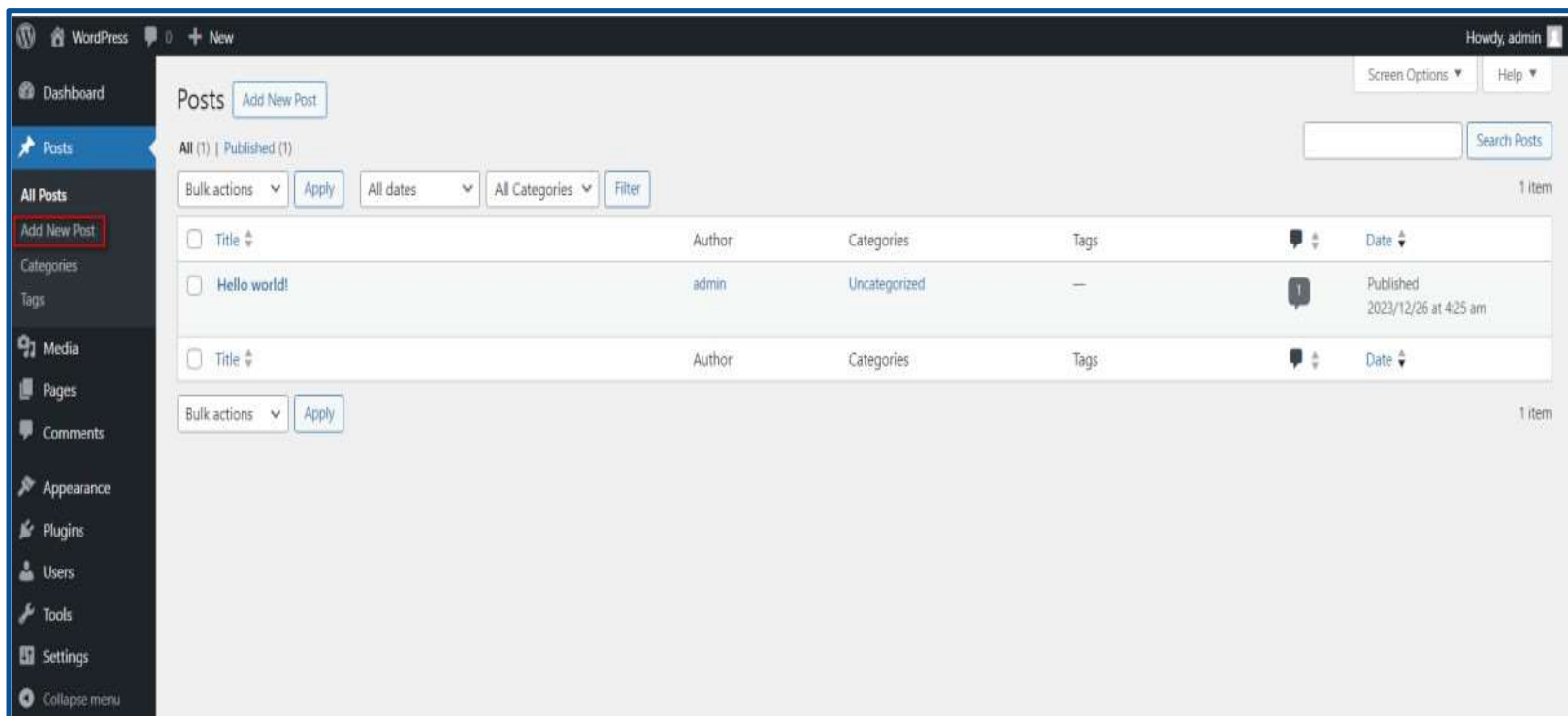


Figure 2.32: WordPress Welcome Page

Fill in Test Post/Page Details: Enter a title for the test post or page in the designated field(*Figure 2.34: Post creation Page*).

The image shows two parts of the WordPress interface. The top part is the 'Posts' management screen in the dashboard, and the bottom part is the 'Add New Post' editor.

Top Screenshot (Dashboard):

- Header: WordPress logo, home icon, 'WordPress' text, 0 notifications, and a '+ New' button.
- Left Sidebar: 'Dashboard' (selected), 'Posts', 'All Posts', 'Add Post', 'Categories', 'Tags', 'Media', 'Pages', and 'Comments'.
- Main Content Area: 'Posts' section with an 'Add Post' button. Below it, a filter bar shows 'All (1) | Published (1)'. There are buttons for 'Bulk actions', 'Apply', 'All dates', 'All Categories', and 'Filter'.
- Table:

☐	Title	Author	Categories	Tag
☐	Hello world!	Admin	Uncategorized	—
☐	Title	Author	Categories	Tag

Below the table are 'Bulk actions' and 'Apply' buttons.

Bottom Screenshot (Post Editor):

- Header: '+', undo, redo, and a menu icon. The title field contains 'New title - Post' and 'Ctrl+K'.
- Buttons: 'Save draft', 'Publish', and a 'Post' button.
- Main Content Area: A large text area with 'Add title' and 'Type / to choose a block'. A block inserter is open, showing options for Paragraph, Image, Video, and Audio. A 'Welcome to the editor' modal is also visible.
- Right Sidebar: 'Post' and 'Block' tabs. The 'Post' tab is active, showing fields for 'No title', 'Set featured image', 'Add an external image', 'Last edited a second ago', 'Status' (Draft), 'Publish' (Immediately), 'Slug' (a), 'Author' (Admin), 'Template' (Single Posts), 'Discussion' (Open), 'Format' (Standard), and 'Categories'.



Figure 2.35: Post Sample Page